Energy 361 (2026) 142089

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Multi-timescale optimal scheduling of an electrolytic aluminum industrial park: coupling of heterogeneous production processes and transmission-distribution dual risk

Yulong Yang [a,*], Songnan Wang [a], Gangui Yan [a], Gang Mu [a], Zhe Chen [b]

[a] Key Laboratory of Modern Power System Simulation and Control & Renewable Energy Technology, Ministry of Education (Northeast Electric Power University), Jilin, 132012, China
[b] Department of Energy Technology, Aalborg University, Aalborg, 9220, Denmark

ARTICLE INFO

Keywords:
Electrolytic aluminum load (EAL)
Coordinated regulation
Peak-shaving ancillary service
Dual uncertainty
Conditional value-at-risk (CVaR)

ABSTRACT

With the increasing penetration of renewable energy into power systems, the electrolytic aluminum load (EAL), as an energy-intensive industrial consumer, exhibits substantial regulation potential and can function as a key flexible resource for grid interaction. In this study, we aimed to characterize the heterogeneous regulation capabilities of the EAL across its three core production processes—electrolysis, feeding, and aluminum tapping—and to enable coordinated dispatch between distribution-side flexible resources and the transmission system under dual uncertainties stemming from captive wind power fluctuations and peak-shaving market price volatility. First, a refined multi-process regulation model is developed to systematically capture the differences among the three processes in their operational mechanisms, response speeds, and adjustable capacities. Second, to explicitly address uncertainties from both the transmission and distribution sides, a temporal conditional value-at-risk (CVaR)-based peak-shaving capacity correction model is proposed to dynamically revise the available peak-shaving capacity. Finally, the three production processes are allocated to the day-ahead, intra-day, and real-time scheduling stages according to their distinct regulation characteristics and are jointly coordinated with captive thermal power units, thereby facilitating wind power utilization and grid peak-shaving services. Case study results demonstrate that the proposed method significantly enhances wind power utilization and overall economic performance while improving the reliability of EAL participation in grid operations.

Nomenclature

Sets	
t	index of scheduling periods
z	index of electrolytic cell groups
d	index of cell within the group
m	index of vacuum tapping machines
i	index of time intervals
Parameters	
N	total number of cells
T_N	rated electrolyte temperature, °C
E	electrochemical reaction voltage, V
N_λ	maximum number of regulation actions
η_N	rated current efficiency
T^{keep}	minimum stabilization time after regulation
A	heat transfer area of the cell body, m^2
h_{conv}	convective heat transfer coefficient, $MW \cdot m^{-2} \cdot °C^{-1}$

(*continued on next column*)

(*continued*)

C_0	radiation coefficient of a black body, $MW \cdot m^{-2} \cdot K^{-1}$
ε	emissivity of the radiating surface
φ	radiation angle factor between adjacent surfaces
$m_{z,d}^{Al_init}$	initial molten aluminum content, kg
m_e/m_s	masses of electrolyte/cell shell, kg
c_e/c_s	specific heat capacities of electrolyte/cell shell, J/(kg·°C)
E^{jl}/E^{cl}	power consumption coefficient of feeding/aluminum tapping process, MW/kg
$a_i/b_i/c_i$	cost parameters of thermal power units
c_{st}	start-up cost of thermal power unit, CNY/start
$c_{Al}/c_{ele}/c_t^{pe}/c_{load}$	aluminum revenue coefficient, time-of-use electricity price, peak-shaving/grid compensation for industrial load, CNY/MW
λ_w	wind curtailment cost, CNY
I_{max}/I_{min}	upper/lower limits of series current, A
T_{max}/T_{min}	upper/lower limits of electrolyte temperature, °C
$m_{max}^{Al}/m_{min}^{Al}$	upper/lower limits of molten aluminum content, kg

(*continued on next page*)

* Corresponding author.
E-mail address: yangyulong@neepu.edu.cn (Y. Yang).

https://doi.org/10.1016/j.energy.2026.142089
Received 27 December 2025; Received in revised form 19 June 2026; Accepted 28 July 2026
Available online 3 August 2026
0360-5442/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(*continued*)

$u^p_{\max}$/ $u^p_{\min}$	upper/lower limits of alumina conveying rate, kg
$q_{\max}$/ $q_{\min}$	upper/lower limits of daily tapping volume per cell, kg
$P^{dj}_{\max}$/ $P^{dj}_{\min}$	upper/lower power limits of electrolysis process, MW
$P^{G}_{\max}$/ $P^{G}_{\min}$	upper/lower output limits of thermal power unit, MW
P^{dj}_{up}/ P^{dj}_{down}	upward/downward ramp limits of electrolysis process, MW
P^{il}_{up}/ P^{il}_{down}	upward/downward ramp limits of feeding process, MW
P^{G}_{up}/P^{G}_{down}	upward/downward ramp limits of thermal power unit, MW
Variables	
I_t	series current, A
U_t	cell voltage, V
η_t	actual current efficiency
Q^{conv}_t	convective heat loss, MWh
Q^{rad}_t	radiative heat loss, MWh
q^{rad}_t	radiative heat flux density, MW·m^{-2}
T^s_t/ T^e_t/ T^0_t	temperature of electrolyte/cell shell/ambient air, °C
λ^{up}_t/ λ^{keep}_t/ λ^{down}_t	power status of electrolysis process (binary variable)
u^p_t	alumina conveying rate, kg
u^f_t	alumina consumption/molten aluminum generation rate, kg
$m^{Al_2O_3}_t$/ $m^{Al}_{z,d,t}$	alumina/molten aluminum content, kg
$q_{z,d,t}$	aluminum tapping volume, kg
$x_{z,d,t}$	vacuum tapping operation status (binary variable)
$y_{m,z,d,t}$	service status of vacuum tapping machine (binary variable)
$s_{i,t}$	start-up/shut-down of thermal power unit (binary variable)
K^w_t/ K^a_t	wind power forecast error rate/awarded capacity violation rate
V^w_t/ V^a_t	wind power forecast error/awarded capacity violation risk cost, CNY
P^{dj}_t/ P^{il}_t/ P^{cl}_t	power consumption of electrolysis/feeding/aluminum tapping process, MW
P^{Al}_t	total power consumption of EAL, MW
$P^{G}_{i,t}$	output of thermal power unit, MW
P^{wf}_t/ P^{w}_t	forecast/actual wind power output, MW
P^{bid}_t/ P^{awd}_t/ P^{act}_t	bid/awarded/intra-day actual adjustable capacity, MW

1. Introduction

The increasing penetration of renewable energy sources (RES) has displaced conventional thermal generation, resulting in a growing scarcity of system regulation capability [1]. Moreover, the inherent stochasticity and intermittency of RES intensify supply–demand imbalances, which have led to widespread wind and solar curtailment [2] and have exposed the limitations of conventional supply-side resources. As a result, mobilizing flexible resources on the distribution side has become essential for the transition from the traditional generation-follows-load paradigm toward a more interactive source–grid–load framework [3]. In this context, energy-intensive industrial loads—particularly the electrolytic aluminum load (EAL)—have emerged as promising candidates for enhancing grid interaction and renewable energy integration. This role is primarily attributed to their large installed capacity and high controllability [4], as well as to their urgent need for a low-carbon transition [5]. Consequently, they have become key participants in grid-interactive regulation and renewable energy absorption [6].

Against this backdrop, modeling the characteristics and operational strategies of EAL participation in grid regulation has become a research focus. Analyzing the electro-thermal coupling of the electrolysis process, existing studies have established a dynamic model linking electrolyte temperature and current [7]. Ref. [8] further incorporated production variables such as current, temperature, and aluminum output, providing a more comprehensive characterization of the EAL regulation capability. Other studies have focused on its regulation capability to propose scheduling strategies for participating in ancillary service markets for peak-shaving [9] and frequency regulation [10], as well as to propose source-load coordinated scheduling methods at the industrial park level [11]. These studies demonstrate that EAL participation in regulation can enhance power system flexibility.

Existing studies mainly focused on the electrolysis process itself. In practical engineering, this process relies primarily on on-load tap-changing transformers for coarse adjustment of the series current [12] and on saturated reactors for fine adjustment [13]; the coordination of these two devices enables effective regulation of the cell current. However, electrolytic aluminum production involves a tightly coupled feeding–electrolysis–tapping chain: molten aluminum is continuously generated via electrochemical reactions under direct current [14], alumina must be replenished periodically to maintain the in-cell material balance [15], and molten aluminum is extracted through tapping operations [16], with the coupling relationships among the three processes illustrated in Fig. 1. The heterogeneous regulation characteristics and coordinated flexibility of feeding and tapping remain underexplored, which constrains the exploitable regulation capacity of EAL.

Regarding optimal scheduling under uncertainty, existing literature has conducted research from various perspectives. Distributionally robust optimization (DRO) has been applied to address photovoltaic output fluctuations in building-integrated photovoltaic (BIPV) systems [17], uncertainties in integrated energy systems (IES) involving carbon capture and power-to-hydrogen [18], and load-side fluctuations in electricity-heat-hydrogen IES [19]. Conditional value-at-risk (CVaR) has been introduced to quantify the economic risk of park-level integrated energy systems (PIES) posed by uncertainties in natural gas and electricity prices and their correlations [20], as well as to address the dispatch deviation risk of electric-hydrogen integrated energy stations (EHES) under source-load uncertainties in energy and reserve service markets [21]. Ref. [22] proposed deep learning (DL) to characterize the dual uncertainties in a hydro-wind-solar complementary system and constructs a short-term stochastic optimization (SO) model based on scenario reduction. Ref. [23] proposed a robust-stochastic optimization model for industrial park loads that accounts for multiple coupled uncertainties, including renewable energy output, dynamic carbon intensity, and electricity price.

However, renewable energy output exhibits significant diurnal fluctuation patterns [24], and the distribution of its forecast error mainly depends on the point forecast value and the time period [25]. The uncertainty in rival bidding prices in the peak-shaving market, which is often influenced by unit status and market strategies [26], leads to temporal variations in the awarded capacity. As an industrial load on the distribution network side, the electrolytic aluminum industrial park faces two types of temporal uncertainties when participating in grid ancillary services: the fluctuation of captive renewable energy output on the distribution side and the uncertainty of rival bidding prices on the transmission side. Although existing studies have made progress in handling uncertainties, the types of uncertainties they address differ fundamentally from the transmission–distribution dual temporal uncertainty faced by the electrolytic aluminum industrial park. Currently, there remains a lack of effective methods capable of uniformly addressing this type of problem.

Multi-timescale optimal scheduling has witnessed significant research interest within power systems. For hydrogen-electricity port systems, Ref. [27] demonstrated that slow-responding resources can stabilize base loads while fast-responding assets mitigate real-time disturbances. Regarding multi-carrier IES, Ref. [28] formulated a multi-time-scale strategy incorporating stepped carbon trading. In the industrial sector, such scheduling is typically achieved by leveraging response-speed disparities between desalination and electro-thermal loads [29], or by exploiting the inherent characteristics of diverse industries such as cement and steel [30] to meet specific operational objectives. Furthermore, Ref. [31] dynamically assigned heterogeneous power-to-heat (P2H) loads to distinct scheduling windows based on their instantaneous states. However, existing literature primarily centers on the complementary coordination between discrete load resources or different states of a single load. A critical gap remains in the granular exploration of regulation characteristics among multiple production processes within a single industrial entity and their adaptive alignment with varying timescales.

Electrolytic aluminum production involves three core processes: electrolysis, feeding, and aluminum tapping. These processes exhibit

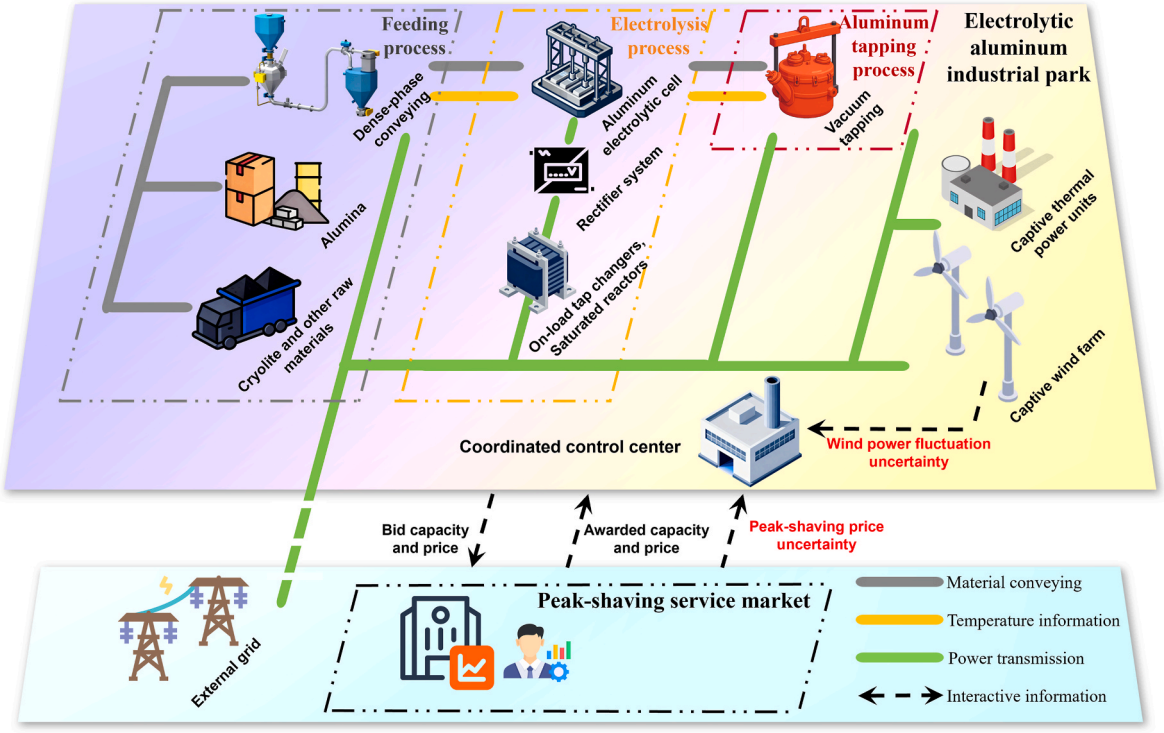


Fig. 1. System framework of the proposed electrolytic aluminum industrial park.

significant differences in regulation capacity, response speed, and duration. This multi-process heterogeneity determines that a single-timescale scheduling method cannot fully exploit their coordinated potential. Ref. [32] proposed a multi-process coordinated power control strategy for electrolytic copper, but it is not applicable to EAL. Ref. [33] constructed a multi-timescale scheduling method that considers the characteristics of the electrolysis process, but completely ignored the differentiated regulation capability of the feeding and aluminum tapping processes and their adaptation to different timescales. The strengths and weaknesses of the relevant literature are summarized in Table 1.

To address the aforementioned issues, the main contributions of this study are summarized as follows:

Table 1
A summary of related Reference.

Research point	Ref.	Research object	Strengths	Weaknesses
Modeling and scheduling of EAL regulation characteristics	[7]	EAL	Electro-thermal dynamic modeling	Neglects feeding/tapping process modeling
	[8]	EAL	Integrated modeling of production variables	Simplified modeling of feeding/tapping characteristics
	[9]	EAL	Collaborative optimization of EAL and deep thermal power peak shaving	Lacks multi-process coordinated regulation
	[10]	EAL	Frequency regulation ancillary service integration	Lacks multi-process coordinated regulation
	[11]	EAL	Park-level source-load coordination	Lacks multi-process coordinated regulation
Uncertainty handling and risk quantification	[17]	BIPV system	DRO for dual source-load uncertainty	Neglects temporal differentiation in RES fluctuation analysis
	[18]	IES	Day-ahead DRO and intraday model predictive control	Neglects temporal differentiation in RES fluctuation analysis
	[19]	IES	DRO with comprehensive norm constraints and generative adversarial network	Neglects temporal differentiation in RES fluctuation analysis
	[20]	PIES	CVaR-based quantification of gas and electricity price risks	Conventional CVaR inapplicable to transmission-distribution dual uncertainty
	[21]	EHES	CVaR for market deviation risk under source-load uncertainty	Conventional CVaR inapplicable to transmission-distribution dual uncertainty
	[22]	Hydro-wind-solar system	SO for dual uncertainty under DL scenarios	Scenario reduction neglects extreme cases
	[23]	Industrial park loads	Robust-stochastic optimization for source-load-price multivariate uncertainty	Imbalance between conservativeness and risk across time periods
Multi-timescale optimal scheduling	[29]	Desalination and electro-thermal loads	Utilization of response rate differences in heterogeneous loads	Neglects inter-process coupling constraints
	[30]	Multiple industrial loads	Utilization of heterogeneous characteristics in industrial loads	Neglects inter-process coupling constraints
	[31]	P2H	Real-time state-based dynamic allocation across time windows	Distinct load characteristics
	[32]	Electrolytic copper load	Multi-process coordination mechanism	Distinct load characteristics
	[33]	EAL	Based on electrolysis process response characteristics	Neglects feeding/tapping heterogeneity and timescale adaptation

(1) This study reconceptualizes EAL flexibility as a heterogeneous multi-process regulation system, rather than as a monolithic adjustable load. By explicitly modeling the feeding and aluminum tapping processes under continuous and discrete dynamic material balance constraints and integrating them with the electrolysis process under electro-thermal coupling, a refined multi-process regulation model is established, revealing regulation capabilities that are overlooked in conventional electrolysis-only models.
(2) Addressing the dual internal and external uncertainties arising from captive wind power fluctuations and peak-shaving market price volatilities, a temporal CVaR-based peak-shaving capacity correction model is developed to dynamically adjust the peak-shaving capacity of the electrolytic aluminum industrial park, ensuring both its economic efficiency and the reliability of grid operation.
(3) Based on the differentiated regulation characteristics—namely, the large-capacity and low-frequency regulation of the electrolysis process, the discrete step-change regulation of the aluminum tapping process, and the rapid fine-tuning of the feeding process—these production processes are allocated across day-ahead, intra-day, and real-time timescales and coordinated with captive thermal power units, thereby enhancing local wind power utilization.

The system framework of the proposed electrolytic aluminum industrial park is shown in Fig. 1. The park consists of a load zone and power generation units. The load zone is the EAL, whose production includes three coupled processes: electrolysis, feeding, and aluminum tapping. The generation units include a captive wind farm and thermal power units, achieving a self-sufficient power supply within the park. The park is connected to the external grid and interacts with the peak-shaving market through a control center to provide peak-shaving ancillary services.

2. Methodology

2.1. Refined regulation model for multi-process in electrolytic aluminum load

To exploit the regulation capability of EAL across multiple production processes, an electro-thermal coupled electrolysis model under practical production constraints, a feeding model with continuous dynamic material balance constraints, and an aluminum tapping model with discrete dynamic material balance constraints are constructed as described below.

2.1.1. Analysis and modeling of the production characteristics and regulation capability of the electrolysis process

The electrolysis process model is constructed under constraints including electro-thermal coupling and reaction-rate limits associated with the material balance of the feeding process.

1) Current-power coupling constraint

$$P_t^{dj} = N(U_t - E)I_t \tag{1}$$

$$P_{\min}^{dj} \le P_t^{dj} \le P_{\max}^{dj} \tag{2}$$

$$I_{\min} \le I_t \le I_{\max} \tag{3}$$

2) Electro-thermal coupling constraint

The electrical energy input into the electrolytic cell is converted into heat for heating the raw materials, denoted as H_h; energy consumption for the electrochemical reaction, denoted as Q_{chem}, and heat dissipation from the electrolyte to the cell shell, denoted as Q_{sur}. The theoretical energy income W_{in} and expenditure W_{out} of the electrolyte within the electrolytic cell over a time period dt are respectively expressed as follows:

$$W_{in} = P_t^{dj} dt \tag{4}$$

$$W_{out} = Q_{sur} + Q_{chem} + H_h \tag{5}$$

The calculation formulas for the above physical quantities are as follows:

$$Q_{sur} = \left(Q_t^{conv} + Q_t^{rad}\right)dt \tag{6}$$

$$Q_t^{conv} = Ah_{conv}\left(T_t^e - T_t^s\right) \tag{7}$$

$$Q_t^{rad} = Aq_t^{rad} = A\varepsilon C_0 \varphi \left[\left(\frac{T_t^e + 273}{100}\right)^4 - \left(\frac{T_t^s + 273}{100}\right)^4\right] \tag{8}$$

The model targets the outer surface of the electrolytic cell shell; therefore, ε is set as 0.8, φ is simplified to 0.5 for other parts [34], and the value of C_0 is 5.67×10^{-6}MW·m^{-2}·K^{-1}.

The sum of Q_{chem} and H_h is the theoretical energy consumption Q_c of the entire electrolysis reaction process:

$$Q_{chem} + H_h = Q_c = 0.3356 \times (5.63 + 0.7 - 4.32(1 - \eta_t))I_t dt \tag{9}$$

where 0.3356 represents the electrochemical equivalent of aluminum, kg/(kA·h); 5.63 V, 0.7 V, and 4.32 V represent the theoretical decomposition voltage, equivalent voltage for heating the raw materials, and equivalent voltage for the exothermic secondary reaction, respectively, expressed in Ref. [35].

Research focusing on electrolytes includes:

$$P_t^{dj} - \frac{Q_{sur}}{dt} - \frac{Q_{chem}}{dt} - \frac{H_h}{dt} = c_e m_e \frac{dT_t^e}{dt} \tag{10}$$

Research focusing on the electrolytic cell shell includes:

$$\frac{Q_{sur}}{dt} - \frac{Q_{sur1}}{dt} = c_s m_s \frac{dT_t^s}{dt} \tag{11}$$

The calculation formulas for Eq. (11) are similar to those in Eq. (6)–(8); for details, see Appendix C, Eq. (C.1)–(C.3). By applying discretization to Eqs. (10) and (11), we obtain:

$$P_t^{dj} - Q_t^{conv} - Q_t^{rad} - Q_t^c = c_e m_e \left(T_{t+1}^e - T_t^e\right) \tag{12}$$

$$\left(Q_t^{conv} + Q_t^{rad}\right) - \left(Q_t^{conv1} + Q_t^{rad1}\right) = c_s m_s \left(T_{t+1}^s - T_t^s\right) \tag{13}$$

3) Electrolyte temperature constraint

$$T_{\min} \le T_t^e \le T_{\max} \tag{14}$$

4) Current efficiency variation constraint

Power regulation of the electrolytic cell induces variations in the cell temperature, which directly influence the current efficiency. Specifically, for every 10 °C decrease in cell temperature, the current efficiency increases by 1.0%–1.5% [36]. The variation in current efficiency is formulated as follows:

$$\eta_t = \eta_N - 1.5\%\left(T_t^e - T_N\right) \tag{15}$$

5) Electrolysis process power ramp constraint

$$P_t^{dj} = P_{t-1}^{dj} + \Delta P_t^{dj} \tag{16}$$

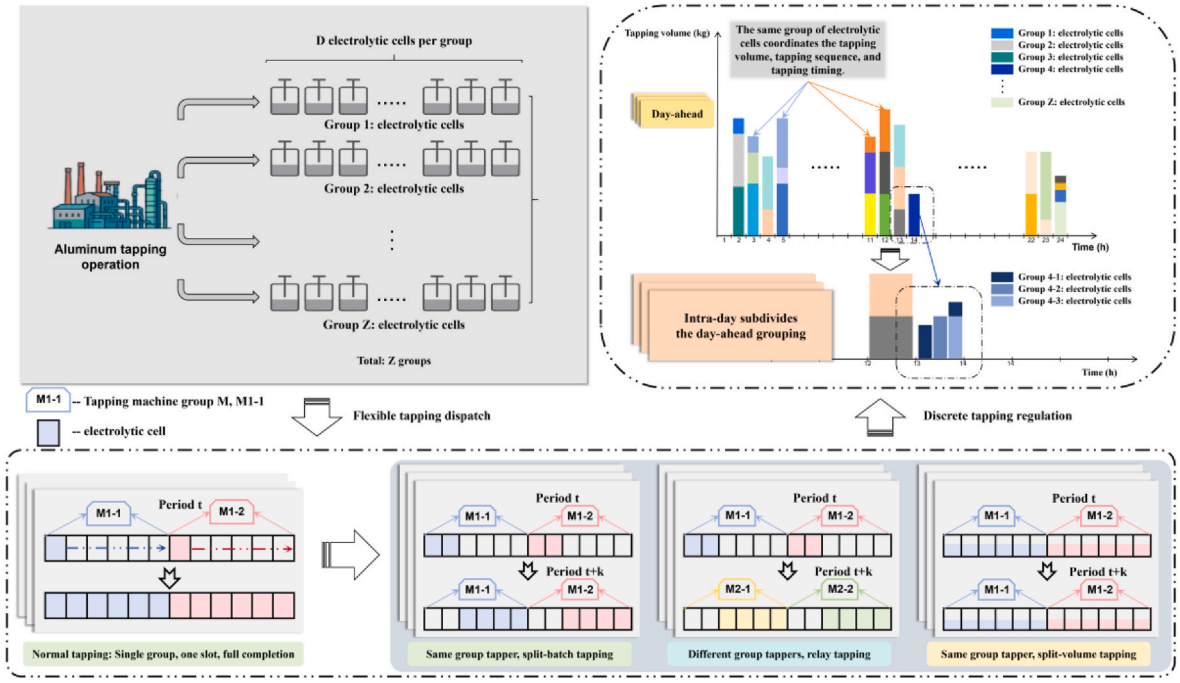


Fig. 2. Aluminum tapping operation mechanism.

$$\lambda_t^{down} P_{down}^{dj} \le \Delta P_t^{dj} \le \lambda_t^{up} P_{up}^{dj} \tag{17}$$

6) Regulation count constraint

$$\sum_{t=1}^{T} \left(\lambda_t^{up} + \lambda_t^{down}\right) \le N_\lambda \tag{18}$$

7) Regulation stability constraint

Given that continuous regulation of the electrolysis process may affect the thermal stability within the electrolytic cell, to ensure this thermal stability, continuous regulation within adjacent time periods is prohibited. The constraints are as follows:

$$\left(\lambda_t^{up} + \lambda_t^{down}\right) + \left(\lambda_{t-1}^{up} + \lambda_{t-1}^{down}\right) \le 1 \tag{19}$$

$$\left(\lambda_{t-1}^{keep} - \lambda_t^{keep}\right)\left(T_{t-1}^{keep} - T_t^{keep}\right) = 1 \tag{20}$$

8) Electrolysis process unique state constraint

$$\lambda_t^{keep} + \lambda_t^{up} + \lambda_t^{down} = 1 \tag{21}$$

2.1.2. Analysis and modeling of the production and regulation characteristics of the feeding process

The feeding process must account for the continuous material balance within the cell under varying electrolysis currents and discrete aluminum tapping schedules. By tracking the alumina concentration, the conveying rate can be adjusted to indirectly regulate power consumption.

Therefore, a detailed feeding process model is required, which considers raw material distribution, consumption rates, and overall material balance.

1) Consumption and conveying rate constraint

$$u_t^f = \frac{1889 \times 0.3356 \times I_t \times \eta_t}{1000} \tag{22}$$

$$u_{\min}^p \le u_t^p \le u_{\max}^p \tag{23}$$

where 1889 represents the alumina consumption per ton of aluminum produced, $kg_{Al_2O_3}/kg_{Al}$.

3) Feeding process material balance constraint

$$m_{t+1}^{Al_2O_3} = m_t^{Al_2O_3} - u_t^f + u_t^p \tag{24}$$

$$m_{\min}^{Al_2O_3} \le m_t^{Al_2O_3} \le m_{\max}^{Al_2O_3} \tag{25}$$

The upper and lower limits $m_{\min}^{Al_2O_3}$ and $m_{\max}^{Al_2O_3}$ of the alumina content in the cell are determined by the alumina concentration. When the concentration is maintained between 1.9% and 2.3%, the resistance exhibits a linear relationship with the concentration, ensuring high current efficiency [37].

4) Power consumption of the feeding process

$$P_t^{jl} = NE^{jl} u_t^p \tag{26}$$

5) Feeding process power ramp constraint

$$P_t^{jl} = P_{t-1}^{jl} + \Delta P_t^{jl} \tag{27}$$

$$P_{down}^{jl} \le \Delta P_t^{jl} \le P_{up}^{jl} \tag{28}$$

2.1.3. Analysis and modeling of the production and regulation characteristics of the aluminum tapping process

Molten aluminum is generated continuously within the cell and withdrawn through discrete tapping operations. Therefore, the tapping process requires coordinated control of tapping volume, tapping frequency, and tapping timing under discrete material-balance constraints.

The proposed aluminum tapping operation mechanism is illustrated in Fig. 2. In the day-ahead stage, a total of N electrolytic cells are partitioned into Z groups, each consisting of D cells. In addition, M groups of vacuum tapping machines are assigned to perform aluminum extraction. Under the premise that all cells in the same group follow an identical schedule, three tapping modes are incorporated as needed: same-group

split-batch, different-group split-cell relay, and same-group split-volume. Combinatorial optimization of these modes enables coordinated adjustment of tapping volume, sequence, and timing across groups, achieving optimal scheduling.

In the intra-day stage, the aforementioned Z groups are further subdivided into 3 Z subgroups, each containing $D/3$ electrolytic cells. While maintaining the original operating rules unchanged, this refined grouping enables regulation capability to be exploited at finer time scales. Within each subgroup, different tapping volumes can be scheduled across different time periods for aluminum tapping operations, subject to tapping volume constraints and the discrete material balance of molten aluminum within each electrolytic cell.

Under this mechanism, the modeling for the aluminum tapping process is as follows:

1) Tapping operation-time window constraint

The scheduling of aluminum tapping operations must align with the established shift schedules (T_1、T_2、T_3), thereby preventing the introduction of additional operational complexity and safety risks due to excessively dispersed tapping times.

$$\sum_{t\notin T_1\cup T_2\cup T_3}^{T} x_{z,d,t}=0 \tag{29}$$

2) Single-electrolytic-cell tapping volume constraint

Within the electrolysis workshop, each electrolytic cell continuously undergoes electrolysis reactions to produce molten aluminum. To maintain the dynamic balance inside the cell, both the daily tapping volume and the single-tapping amount per operation must remain within the specified tapping indication range.

$$0\le q_{z,d,t}\le x_{z,d,t}q_{\max} \tag{30}$$

$$q_{\min}\le\sum_{t}^{T}x_{z,d,t}q_{z,d,t}\le q_{\max} \tag{31}$$

3) Vacuum-tapping-machine scheduling constraint

To prevent temporal conflicts of vacuum tapping machines caused by cross-group services within the same period and avoid unreasonable no-assignment scenarios, the following constraints are established:

$$\sum_{z=1}^{Z}y_{m,z,t}\le 1 \tag{32}$$

$$x_{z,t}=\sum_{m=1}^{M}y_{m,z,t} \tag{33}$$

4) Aluminum tapping process material balance constraint

The molten aluminum level is a key technical parameter of the electrolytic cell [38] and must be maintained within the normal operating range.

$$m_{z,d,t}^{Al}=m_{z,d}^{Al_init}+\sum_{\tau=1}^{t}u_{\tau}^{f}-\sum_{\tau=1}^{t}q_{z,d,\tau} \tag{34}$$

$$m_{\min}^{Al}\le m_{z,d,t}^{Al}\le m_{\max}^{Al} \tag{35}$$

5) Power consumption of the aluminum tapping process

$$P_t^{cl}=\sum_{z=1}^{Z}\sum_{d=1}^{D}P_{z,d,t}^{cl}=\sum_{z=1}^{Z}\sum_{d=1}^{D}E^{cl}q_{z,d,t} \tag{36}$$

2.2. Risk quantification and assessment of wind power prediction error and peak-shaving price fluctuations

To address the transmission-distribution dual uncertainty faced by the industrial park, the wind power forecast error rate K_t^w and awarded the capacity violation rate K_t^a are constructed to reflect the temporal differences of the two types of uncertainty. A temporal CVaR-based peak-shaving capacity correction model is established to calculate the risk value for each period, yielding the dual uncertainty correction coefficients and achieving quantitative assessment of both risks. The details are as follows:

2.2.1. Analysis of temporal differences in wind power prediction errors

First, the ratio of the difference between historical forecasted wind power P_t^{wf} and historical actual wind power $P_t^{w_act}$ to the installed capacity P^{cap} is defined as the wind power forecast error ΔP_t^w:

$$\Delta P_t^w=\frac{P_t^{w_act}-P_t^{wf}}{P^{cap}} \tag{37}$$

Second, a day is divided into N_1 time intervals (each period contains T_i hours). For each interval, the forecast error rate is averaged within the interval to obtain ΔP_t^w. Then, the average value of ΔP_t^w over N_w groups of historical data is approximated, and finally, the time-interval characteristic vector of the wind power forecast error rate K_t^w is obtained:

$$\Delta P_i^w=\frac{1}{T_i}\sum_{t=1}^{T_i}\Delta P_t^w \tag{38}$$

$$K_i^w=\frac{1}{N_w}\sum_{n=1}^{N_w}\Delta P_{i,n}^w \tag{39}$$

$$K_t^w=\left\{K_1^w,K_2^w,\ldots,K_{N1}^w\right\} \tag{40}$$

2.2.2. Analysis of temporal price variations in the peak-shaving market

The peak-shaving market operates under a two-stage mechanism: day-ahead bid pre-clearing and intra-day formal clearing [39]. Each 15-min interval of operating day D constitutes a trading period. Before 11:00 on day D-1, the dispatch agency publishes the peak-shaving capacity demand D_t and benchmark compensation price c_t^{base} for each period; market participants submit their bids by 12:00. The agency then pre-clears the market in ascending order of bid prices to determine preliminary awarded capacities. From 00:00 of day D, it performs rolling revisions of the demand every 15 min and re-clears the market using the same bids to obtain the final results.

2.2.2.1. Market bidding model. The electrolytic aluminum industrial park submits its bid at price c_t^{base}, with a bid capacity of P_t^{bid}. To simplify the representation, the competitor submits a bid at price r_t, with a declared capacity of P_t^{th}. The awarded capacities are P_t^{awd} and $P_t^{th.awd}$, respectively.

The market clearing aims to minimize the total cost of peak-shaving service:

$$\min\sum_{t=1}^{T}\left(c_t^{base}P_t^{awd}+r_tP_t^{th,awd}\right) \tag{41}$$

$$P_t^{awd}+P_t^{th,awd}=D_t \tag{42}$$

The marginal clearing price c_t^{pe} is defined as the bid price of the last awarded bidder, and all awarded bidders are settled at this price [40]:

$$c_t^{pe} = \max\left\{c_t^{base} \cdot 1_{P_t^{awd}>0}, r_t \cdot 1_{P_t^{th,awd}>0}\right\} \tag{43}$$

2.2.2.2. Simulation of uncertainty in historical rival bidding prices. The uncertainty in the bid price of the rival is represented by a probability distribution. Assume that its bid price $\tilde{r}_{j,k}$ in the k-th segment ($k = 1,2,\ldots, K$, where K is the maximum number of bid segments) follows a normal distribution, with the probability density function given by:

$$p(\tilde{r}_{j,k}) = \frac{1}{\sqrt{2\pi}\delta_{j,k}} \exp\left(-\frac{\left(\tilde{r}_{j,k} - \mu_{j,k}\right)^2}{2\delta_{j,k}^2}\right) \tag{44}$$

where $\delta_{j,k} = \kappa\mu_{j,k}$ is the standard deviation, and κ is the fluctuation coefficient that controls the degree of price volatility. The bid prices in different segments are mutually independent. The Monte Carlo method is used to sample $\tilde{r}_{j,k}$ N_s times. Each sampling yields a set of bid prices $(r_1, r_2, \ldots, r_K)$, forming a bid curve for one operating day. This generates a set of N_s bid scenarios, denoted as $S = \{s_1, s_2, \ldots, s_{N_s}\}$.

Then, based on the model established in Section 2.2.2.1, the awarded capacities $P_t^{adj.cl}$ of the EAL industrial park in each period under different scenarios S are obtained.

2.2.2.3. Data processing. The ratio of the difference between P_t^{awd} and P_t^{bid} to P_t^{bid} is defined as the awarded capacity deviation ΔP_t^{awd}:

$$\Delta P_t^{awd} = \frac{P_t^{awd} - P_t^{bid}}{P^{bid}} \tag{45}$$

The remaining derivation follows the same procedure as that expressed in Eq. (38)–(40). By processing historical bid data in segments, the time-interval average characteristic vector $\{K_1^a, K_2^a, \ldots, K_{N2}^a\}$ of the awarded capacity violation rate K_t^a is obtained.

2.2.3. Temporal CVaR–based peak-shaving capacity correction model

Based on the standard mathematical model of CVaR (Appendix C, Eq. (C.4)–(C.5)), a CVaR-based risk assessment model for wind power prediction errors is constructed to characterize the uncertainty risk of losses caused by these errors. The risk function is defined as follows:

$$f\left(P_i^{wf}, K_i^w\right) = \lambda_w P_i^{wf} K_i^w \tag{46}$$

Wind power risk cost $V\left(P_i^{wf}, a_i^w\right)$:

$$V\left(P_i^{wf}, a\right) = a + \frac{1}{1-\xi}\int \left(f\left(P_i^{wf}, K_i^w\right) - a\right)^+ \rho_i(K_i^w) dK_i^w \tag{47}$$

$$\left(f\left(P_i^{wf}, K_i^w\right) - a\right)^+ = \max\left\{f\left(P_i^{wf}, K_i^w\right) - a, 0\right\} \tag{48}$$

The discretized transformation of the risk cost yields:

$$V_i^w = V\left(P_i^{wf}, a\right) = a + \frac{1}{q(1-\xi)}\sum_{m=1}^{q}\left(f\left(P_i^{wf}, K_i^w\right) - a\right) \tag{49}$$

To characterize the uncertainty risk of awarded capacity violation caused by peak-shaving price fluctuations, the risk function is defined as follows:

$$f\left(P_i^{bid}, K_i^a\right) = c_i^{pe} P_i^{bid} K_i^a \tag{50}$$

The discretized transformation of the risk cost yields:

$$V_i^a = V\left(P_i^{bid}, a\right) = a + \frac{1}{q(1-\xi)}\sum_{m=1}^{q}\left(f\left(P_i^{bid}, K_i^a\right) - a\right) \tag{51}$$

Introducing two risk sensitivity coefficients θ_w and θ_a, the risk costs V_i^w and V_i^a obtained for each time interval i are assigned to each scheduling period t within that interval. The corrected adjustable capacity is expressed as follows:

$$P_t^{adj} = P_t^{av}\left(1 - \theta_w V_t^w\right)\left(1 - \theta_a V_t^a\right) \tag{52}$$

2.3. Multi-timescale optimal scheduling for electrolytic aluminum industrial parks with coordinated wind power consumption and peak-shaving services

2.3.1. Day-ahead scheduling stage

(i) Objective function

In the day-ahead stage, to minimize the total operating cost of the electrolytic aluminum industrial park and maximize its revenue, a weighted objective function model is developed by assigning specific weights to each objective. The total operating cost of the EAL industrial park primarily includes the coal consumption cost C_G of thermal power units, the start-up and shut-down cost C_{st} of thermal power units, and the electricity purchase cost C_{ele}. To optimize wind power consumption, we incorporate the wind curtailment penalty cost C_w into the system cost. The electrolytic aluminum industrial park provides peak-shaving service to the main grid by reducing its own load and releasing captive thermal power and wind power to the main grid. While participating in the peak-shaving market to obtain revenue, the industrial park can also receive load regulation compensation from the grid under the demand response mechanism [41]. Therefore, the total revenue of the electrolytic aluminum industrial park comprises the aluminum production revenue C_{Al}, peak-shaving revenue C_{pe}, and load regulation compensation $C_{compensate}$. The objective function is formulated as follows:

$$\begin{cases} \min \quad (wf_1 - (1-w)f_2) \\ f_1 = C_G + C_{st} + C_w + C_{ele} \\ f_2 = C_{Al} + C_{compensate} + C_{pe} \\ C_G = \sum_{i=1}^{N_G}\sum_{t=1}^{24} s_{i,t}\left(a_i\left(P_{i,t}^G\right)^2 + b_i P_{i,t}^G + c_i\right) \\ C_{st} = \sum_{i=1}^{N_G}\sum_{t=1}^{23} c_{st} s_{i,t+1}\left(1 - s_{i,t}\right) \\ C_w = \sum_{t=1}^{T}\lambda_w\left(P_t^w - P_t^{wf}\right) \\ C_{ele} = \sum_{t=1}^{T} c_t^{ele} P_t^e \\ C_{Al} = \sum_{t=1}^{T} c_{Al} N P_t^{Al} \\ C_{compensate} = \sum_{t=1}^{T} c_{load}\left|P_t^{Al}\right| \\ C_{pe} = \sum_{t=1}^{T} c_t^{pe} P_t^{awd} \end{cases} \tag{53}$$

(ii) Constraints

The constraints primarily include the overall system power balance constraint, thermal power unit output constraints, wind power output constraints, and remaining capacity regulation constraints. The specific mathematical descriptions of each constraint are as follows:

Overall system power balance constraint

$$\sum_{i=1}^{N_G} s_{i,t} P_{i,t}^G + P_t^w + P_t^e = P_t^{Al} + P_t^{awd} \tag{54}$$

$$P_t^{Al} = P_t^{dj} + P_t^{jl} + P_t^{cl} \tag{55}$$

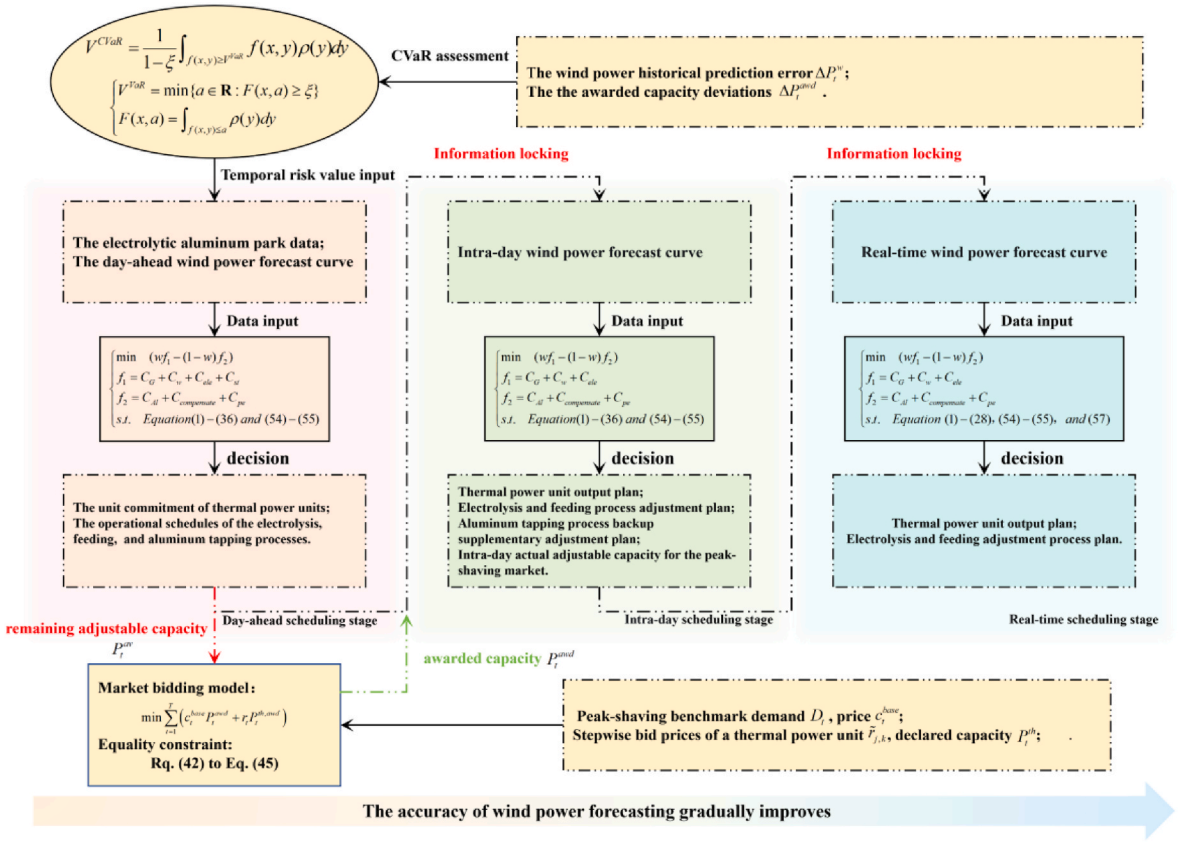


Fig. 3. Solving flowchart of the multi-time scale optimization scheduling model.

The constraints of thermal power units and wind power output are given in Appendix C, Eq. (C.6)–(C.11).

2.3.2. Intra-day scheduling stage

Considering the deviations between day-ahead and intra-day wind power forecast curves and based on the day-ahead unit commitment plan, the intra-day stage leverages the coordinated scheduling mechanism among the three processes, together with the regulation of thermal power units, to accommodate wind power forecast deviations and achieve more accurate optimal scheduling. Unlike the day-ahead stage, this stage does not account for unit start-up/shutdown costs. The compact form of the intra-day optimization scheduling model can be expressed as follows:

$$\begin{cases} \min \quad (wf_1 - (1-w)f_2) \\ f_1 = C_G + C_w + C_{ele} \\ f_2 = C_{Al} + C_{compensate} + C_{pe} \\ s.t. \quad Equation\ (1) - (36)\ and\ (54) - (55) \end{cases} \tag{56}$$

2.3.3. Real-time scheduling stage

In the real-time stage, the regulation capacity of the EAL is constituted solely by the electrolysis process and the feeding process. These two processes are responsible for regulation at the real-time scale, while the aluminum tapping process maintains the scheduling plan from the intra-day stage.

$$\Delta P_t^{Al} = \Delta P_t^{dj} + \Delta P_t^{il} \tag{57}$$

The real-time optimization scheduling model in compact form can be represented as follows:

$$\begin{cases} \min \quad (wf_1 - (1-w)f_2) \\ f_1 = C_G + C_w + C_{ele} \\ f_2 = C_{Al} + C_{compensate} + C_{pe} \\ s.t. \quad Equation\ (1) - (28), (54) - (55), and\ (57) \end{cases} \tag{58}$$

2.3.4. Solution process of the multi-timescale optimization scheduling model

The solution flowchart of the multi-timescale optimization scheduling model is shown in Fig. 3, and the detailed procedure is described as follows:

1) Day-ahead scheduling stage.

The day-ahead schedule is established 24 h in advance with a 1-h temporal resolution. Initially, the day-ahead wind power forecast curve and the EAL park data are incorporated into the day-ahead scheduling model to optimize thermal unit commitment and the operational schedules of the electrolysis, feeding, and aluminum tapping processes. Subsequently, the temporal CVaR-based peak-shaving capacity correction model is applied to adjust the remaining adjustable capacity; the resulting capacity is then integrated into the market bidding model to obtain the day-ahead awarded capacity. Finally, the day-ahead scheduling results, together with the awarded capacity information, are transmitted to the intra-day scheduling stage.

2) Intra-day scheduling stage.

This stage is updated every 15 min, with re-optimization performed over a rolling 4-h horizon. Based on inputs such as the intra-day wind power forecast curve and the day-ahead optimization results, the day-ahead plan is adjusted to accommodate wind power deviations, and the actual peak-shaving capacity is cleared. The final intra-day schedule is then formulated and conveyed to the real-time optimization stage.

3) Real-time scheduling stage.

This stage operates on a 5-min update cycle, with re-optimization performed over a rolling 15-min horizon. Using real-time wind power

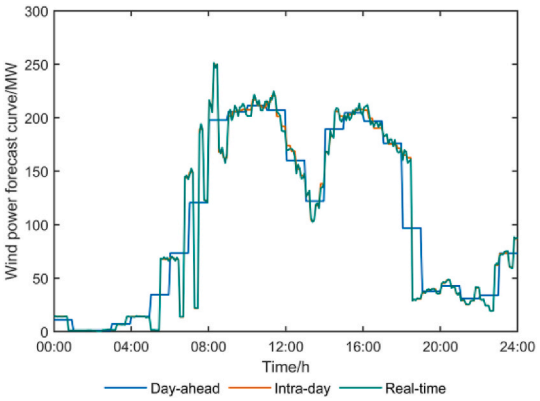


Fig. 4. Wind power forecast curve.

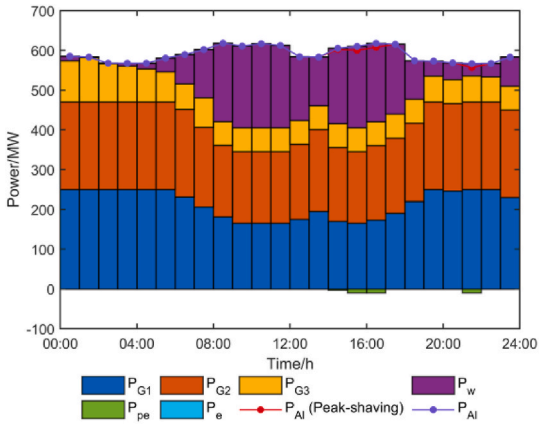


Fig. 5. Day-ahead scheduling balance.

Table 2
Parameters of the aluminum smelting load.

Parameters of the electrolytic aluminum load	Value
Rated current of the electrolytic cell/ KA	500
Number of electrolytic cells in series	288
Upper limit of electrolyte temperature/ ° C	960
Lower limit of electrolyte temperature/ ° C	940
Upper current limit of the electrolytic cell/ KA	510
Lower current limit of the electrolytic cell/ KA	480
Rated current efficiency	92%
Production revenue/ (CNY /t)	15000
Upper limit of tapping volume per electrolytic cell/ kg	2200
Lower limit of tapping volume per electrolytic cell/ kg	2000

data in conjunction with the intra-day optimization results, the operating power of the feeding and electrolysis processes is adjusted dynamically.

All variables in the optimization model are either linear or integer, and no product term exists between them, rendering the model a mixed-integer linear programming (MILP) problem. The time resolutions are 1 h for the day-ahead stage, 15 min for the intra-day stage, and 5 min for the real-time stage. The problem can be solved using the CPLEX solver within the MATLAB environment.

2.4. System overview

We considered an electrolytic aluminum industrial park in Xinjiang as a case study. The plant consists of three captive thermal power units and a 300 MW captive wind farm. The parameters of the captive thermal power units are shown in Appendix B, Table B1. The wind power forecast curve is shown in Fig. 4, and the parameters of the EAL are listed in Table 2. The wind curtailment and compensation costs of the system for industrial load regulation are 200 CNY/ MWh and 150 CNY/ MWh, respectively [42].

For the peak-shaving market, we assumed that the electrolytic aluminum industrial park and a thermal power unit act as peak-shaving capacity providers, and their awarded capacities are determined through competitive bidding [43]. The peak-shaving capacity demand and benchmark compensation price in the peak-shaving market are provided in Appendix B, Table B2; the stepwise bidding prices of the competing thermal power unit are provided in Appendix B, Table B3.

To validate the effectiveness of the proposed method, the following three distinct operational scenarios are established for comparative analysis:

Scenario 1: Only the regulation characteristics of the electrolysis process are considered. The other production processes operate according to the conventional scheduling plan, while the electrolysis process participates in multi-time-scale optimal scheduling based on its production characteristics.

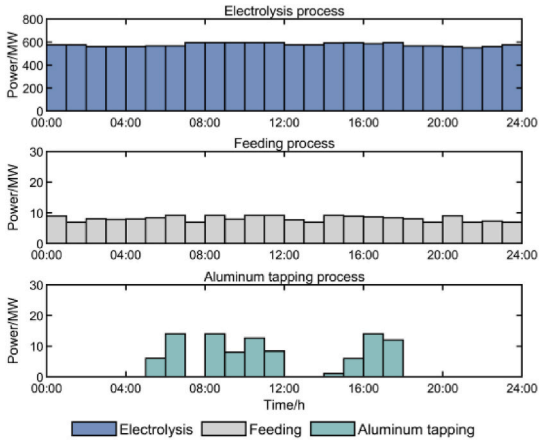


Fig. 6. Day-ahead power consumption of the three processes in the electrolytic aluminum load.

Scenario 2: All three processes of the EAL participate in multi-time-scale optimal scheduling, exploiting the regulation characteristics of the multi-process in a coordinated manner.

Scenario 3: Building upon Scenario 2, the temporal CVaR-based peak-shaving capacity correction model is employed to quantitatively analyze the risks from the dual uncertainties and correct the peak-shaving capacity, followed by multi-time-scale optimal scheduling.

3. Case study

3.1. Regulation capability analysis

The day-ahead scheduling results for Scenario 3 are presented in Figs. 5 and 6. Given that wind power output is concentrated between 06:00 and 19:00, the electrolysis process increases its load consumption above the rated level starting at 08:00, while thermal units G1, G2, and G3 reduce their output to coordinate with the electrolysis process and facilitate large-scale wind power integration. Meanwhile, the feeding process compensates for short-term, high-frequency wind power fluctuations. The aluminum tapping process, scheduled primarily between 07:00–12:00 and 15:00–18:00, provides supplemental regulation by adjusting the tapping volume per unit time and performing discrete operations, thereby filling large, short-duration power gaps that the electrolysis and feeding processes cannot accommodate.

By comparing the load consumption curves before and after participation in peak shaving, it is evident that during peak-shaving demand periods—specifically between 15:00–17:00 and at 22:00—the load is

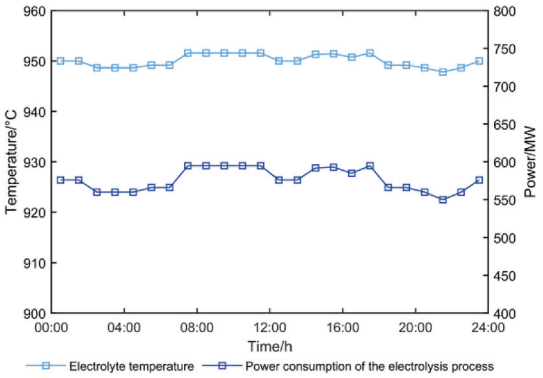


Fig. 7. Electrolytic cell power-temperature regulation characteristics.

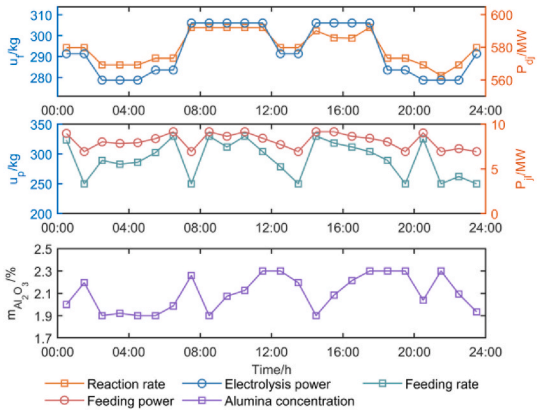


Fig. 8. Analysis of the regulation characteristics of the feeding process.

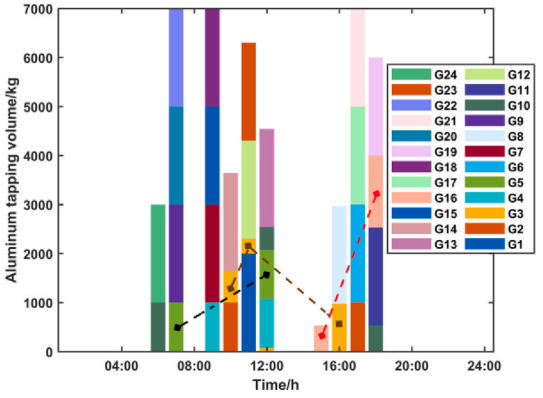


Fig. 9. Aluminum tapping volume per cell group per period.

reduced to provide stable grid support while simultaneously generating additional revenue. Overall, through large-capacity base-load regulation (electrolysis), discrete step-change supplementation (aluminum tapping), and rapid fine-tuning (feeding), the three processes achieve dynamic coordination across multi-timescale scheduling horizons.

As shown in Fig. 7, the temperature of the electrolyzer series fluctuates slightly around 950 °C. The production temperature must be strictly maintained within the range of 940–960 °C, with 950 °C as the rated temperature. The scheduling results stay within the specified constraints, and both power and temperature variations exhibit step-like changes. Each power level satisfies the required minimum holding time per operating point.

Fig. 8 shows that $u_t^p \propto P_t^{jl}$ and $u_t^f \propto P_t^{dj}$. The feeding process dynamically adjusts u_t^p by tracking the power variation of the electrolysis process.

Table 3
Scheduling plan of each tapping machine group from 06:00 to 12:00.

Hour	Machine	Sequence	Hour	Machine	Sequence
6	1	G24(2000) + G10(200)	10	4	G14(1800)
6	2	G10(800)	10	5	G14(200) + G02 (1000) + G03(643)
7	2	G09(1400)	11	1	G01(1643) + G12 (557)
7	3	G09(600) + G20(1600)	11	2	G12(1443) + G23 (757)
7	4	G20(400) + G22(1800)	11	3	G23(1243) + G03 (305)
7	5	G22(200) + G05(1000)	11	5	G01(357)
9	1	G07(2000) + G15(200)	12	3	G13(652)
9	2	G15(1800) + G18(400)	12	4	G13(1348) + G04 (852)
9	3	G18(1600) + G04(600)	12	6	G04(148) + 05(1000) + G10(468) + G03 (76)
9	4	G04(400)			

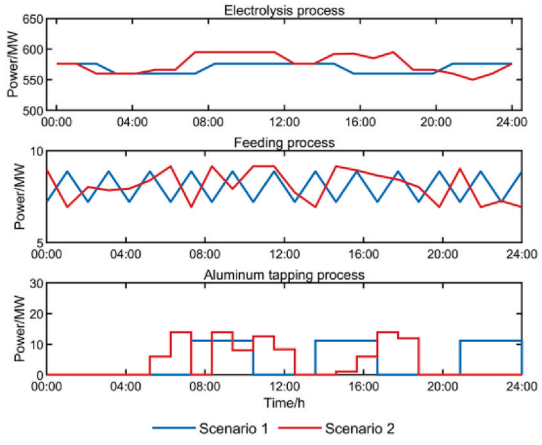


Fig. 10. Comparison of process regulation capability under scenarios 1 and 2.

Under the premise of maintaining the material balance (alumina concentration of 1.9%–2.3%), it leverages its inherent high-frequency, low-amplitude power regulation capability to achieve coupling and coordination with the electrolysis process.

The scheduling results of the tapping process are shown in Fig. 9 and Table 3. Through a combinatorial optimization model for tapping machines and cell groups, the tapping sequence and tapping volume are reasonably allocated (for example, Groups G5, G3, and G16 are tapped in batches across different time periods by multiple tapping machines), achieving discrete step-wise regulation.

3.2. *Analysis of coordinated multi-process regulation for EAL*

As shown in Fig. 10, the electricity consumption variations of the three processes—electrolysis, feeding, and aluminum tapping—are compared for Scenarios 1 and 2. In Scenario 1, only the electrolysis process participates in load regulation, while the feeding and aluminum tapping processes follow their conventional production schedules. Consequently, the electrolysis process exhibits relatively small power variations over the entire scheduling horizon, as its regulation actions must simultaneously accommodate the continuous material supply requirements of the feeding process and the operational constraints of aluminum tapping, thereby significantly constraining its adjustable range.

In contrast, Scenario 2 adopts coordinated regulation across all three processes. Regulation tasks are shared among them, and the electrolysis

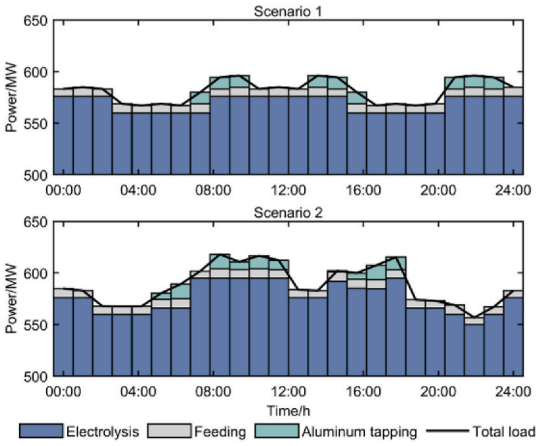


Fig. 11. Total power consumption of the electrolytic aluminum load in Scenarios 1 and 2.

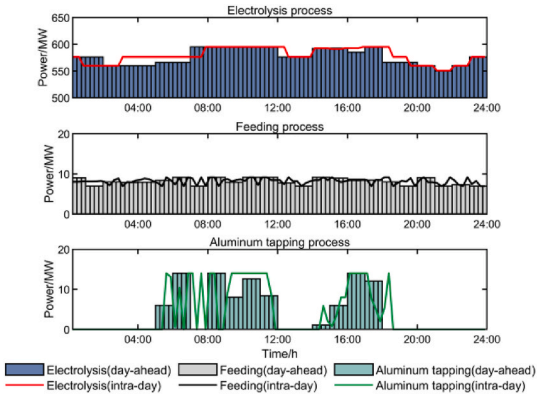


Fig. 12. Comparative analysis of day-ahead and intra-day electrolytic aluminum load regulation capability.

process no longer bears the full burden of load fluctuations alone. The differentiated responses of the three processes are summarized as follows:

(i) Electrolysis process: In Scenario 2, power variations in the electrolysis process are significantly larger than those in Scenario 1, particularly during periods such as 06:00–12:00 and 14:00–18:00, where the electrolysis load exhibits more pronounced increases and decreases. This observation indicates that, under coordinated operation, the electrolysis process can release a greater continuous regulation capacity.

(ii) Feeding process: In Scenario 2, both the frequency and amplitude of power fluctuations increase markedly. During intervals such as 02:00–06:00 and 12:00–18:00, the feeding process undertakes high-frequency, low-magnitude regulation tasks, thereby relieving the electrolysis process from responding to short-term fluctuations.

(iii) Aluminum tapping process: In Scenario 2, the aluminum tapping process exhibits multi-step, discrete power changes during periods such as 06:00–10:00, 15:00–18:00, and after 20:00. This demonstrates how the aluminum tapping process supplements system regulation through discrete operational actions during specific time windows.

Fig. 11 shows the total electricity consumption of the three processes in Scenarios 1 and 2. After the three processes are coordinated in Scenario 2, both the fluctuation characteristics and the regulation magnitude during the wind power peak period (06:00–19:00) are notably superior to those in Scenario 1.

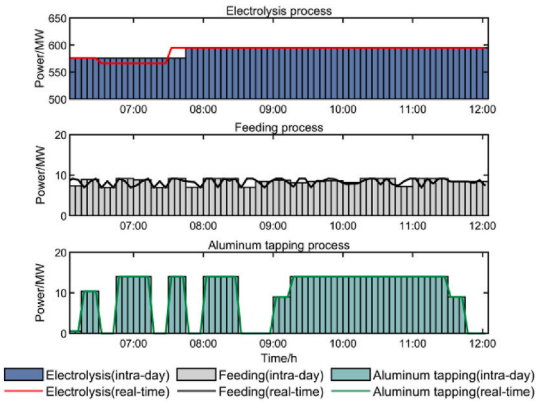


Fig. 13. Comparative analysis of intra-day and real-time electrolytic aluminum load regulation capability.

Fig. 12 compares the day-ahead and intra-day scheduling of the EAL in Scenario 3. Between 03:00 and 09:00, the intra-day wind power forecast deviates significantly from the day-ahead forecast. In response, the electrolysis process rapidly adjusts by increasing its load to counteract these fluctuations in wind power. The aluminum tapping process provides discrete supplementary regulation as needed during the intra-day stage. For example, it performs tapping operations during the 3rd and 4th 15-min intervals at 06:00 and the 2nd and 4th 15-min intervals at 08:00, while halting tapping during the 1st and 3rd 15-min intervals at 07:00. Meanwhile, the feeding process handles frequent, low-magnitude adjustments to track fine-scale wind variations and maintain material balance in the cells.

Overall, the electrolysis process makes larger magnitude load adjustments in the intra-day schedule, the feeding process provides rapid fine-tuning, and the aluminum tapping process offers discrete supplementary regulation. When the combined regulation capacity of these three processes cannot fully absorb the wind power fluctuations, thermal power units step in to provide additional flexible coordination, further facilitating the integration of wind power.

Fig. 13 shows the variation in regulation capability between the intra-day and real-time stages for Scenario 3 during the 06:00–12:00 period. The aluminum tapping process maintains its planned schedule from the intra-day stage without changes in real-time operation. Real-time system regulation is primarily achieved through the coordinated actions of the electrolysis process, the feeding process, and thermal power units: the electrolysis process is down-regulated from 06:00 to 07:00 compared to the intra-day result; the feeding process continuously compensates for fine-scale wind fluctuations with high-frequency, low-magnitude adjustments; and thermal power units manage the remaining power deviations. This coordinated approach facilitates the effective integration of wind power variations while ensuring that all production constraints are met.

The detailed intra-day and real-time scheduling results for each scenario are presented in Appendix A, Figs. A1, A2 and A3, respectively. The output of thermal power units is shown in Fig. A4, and the wind power integration situation is illustrated in Fig. A5.

According to Table 4, compared with Scenario 1, Scenario 3 reduces the wind curtailment cost in the real-time stage from 68.8×10^3 CNY to 15.3×10^3 CNY, increases the integration rate from 84.08% to 93.91%, raises the aluminum product revenue by 168.9×10^3 CNY, and lowers the coal consumption cost by 65.1×10^3 CNY.

3.3. Comparative analysis of capacity correction results under dual uncertainty risks

Fig. 14 validates the proposed CVaR-based model from three perspectives: risk quantification, capacity correction, and scenario comparison..

Table 4
Real-time scheduling results for different scenarios.

	Scenarios 1	Scenarios 2	Scenarios 3
Wind Curtailment Cost/ 10^4CNY	6.88	1.53	1.53
Coal Consumption Cost of Thermal Power Units/ 10^4CNY	381.19	373.73	374.68
Peak-Shaving Market Revenue/ 10^4CNY	0.98	1.17	1.16
Aluminum Product Revenue/ 10^4CNY	1525.90	1541.75	1542.79
Wind Power Consumption Rate/%	84.08	93.91	93.91

(i) In Fig. 14(a), K_t^w peaks during 13:00–16:00, where a high wind power baseline increases relative deviations and V_t^w. K_t^a exhibits high values during 13:00–16:00 and 20:00–24:00, indicating increased revenue loss risk from the peak-shaving market.
(ii) In Fig. 14(b), the corrected adjustable capacity notably reduces during the high-risk periods (13:00–16:00 and 20:00–24:00). Therefore, the model reduces P_t^{bid} in these intervals, trading off some revenue for higher reliability and lower extreme loss risk.
(iii) Comparing Fig. 14(c) and (d), Scenario 2 has much higher day-ahead P_t^{bid} and P_t^{awd} than Scenario 3, but P_t^{act} is similar to that of Scenario 3, leading to a higher default rate. Scenario 3 adopts more robust day-ahead bids: P_t^{awd} lies within the safe region, and P_t^{act} matches P_t^{awd} without deviation or default. As shown in Table 5, the intra-day revenue of Scenario 3 drops by only 200 CNY from the day-ahead value, while that of Scenario 2 drops from 25,800 CNY to 11,300 CNY. Thus, Scenario 3 achieves higher intra-day revenue and zero default at the cost of lower day-ahead revenue, validating the proposed temporal CVaR-based peak-shaving capacity correction model.

3.4. Temporal evolution and correlation analysis of dual uncertainties under multiple scenarios

3.4.1. Comparison of correction effectiveness across multiple scenarios

To analyze the effectiveness of the temporal CVaR correction model under different uncertainty scenarios, four typical scenarios (corresponding to different seasons, with bid price fluctuations realized by adjusting the fluctuation coefficient κ) are selected. The comparison is conducted based on the market demand in Appendix B. Table B2 expanded by a factor of 7, assuming a park bid price of 0.5 times the benchmark compensation price, and the results are shown in Fig. 14(a)–(d).

1) In Fig. 15 (a) (low wind power output with weak fluctuations and small rival bid volatility), the capacity differences before and after correction are minimal, and the correction effect is not significant.
2) In Fig. 15(b)—the most severe dual-uncertainty case, characterized by high wind power output, large day-ahead to intra-day forecast deviations ($\kappa = 0.5$), and strong bid price volatility—large gaps exist before correction between P_t^{awd} and P_t^{act}, as well as between P_t^{bid} and P_t^{awd}, leading to a very high capacity default risk. After correction, these gaps are significantly reduced, especially during the 22:00–24:00 period.
3) In Fig. 15(c) (large temporal wind power variations), P_t^{act} closely matches P_t^{awd} after correction, whereas a notable deviation between the two exists before correction, resulting in default. It should be noted that this outcome is already the result of the EAL's multi-process coordinated regulation smoothing intra-day wind power fluctuations; without such regulation, the default rate would be even higher.
4) In Fig. 15(d) (large temporal variation of market bid prices), the gap between P_t^{bid} and P_t^{awd} is significantly reduced after correction, while P_t^{act} remains almost unaffected. Despite negligible revenue change, the high consistency between P_t^{bid} and P_t^{awd} demonstrates the EAL park's reliable peak-shaving capability, which helps lower market operation risk.

Table 5
Peak-shaving revenue under different scenarios ($\times 10^4$ CNY).

Scenario	Day-ahead	Intra-day
2	2.58	1.13
3	1.20	1.18

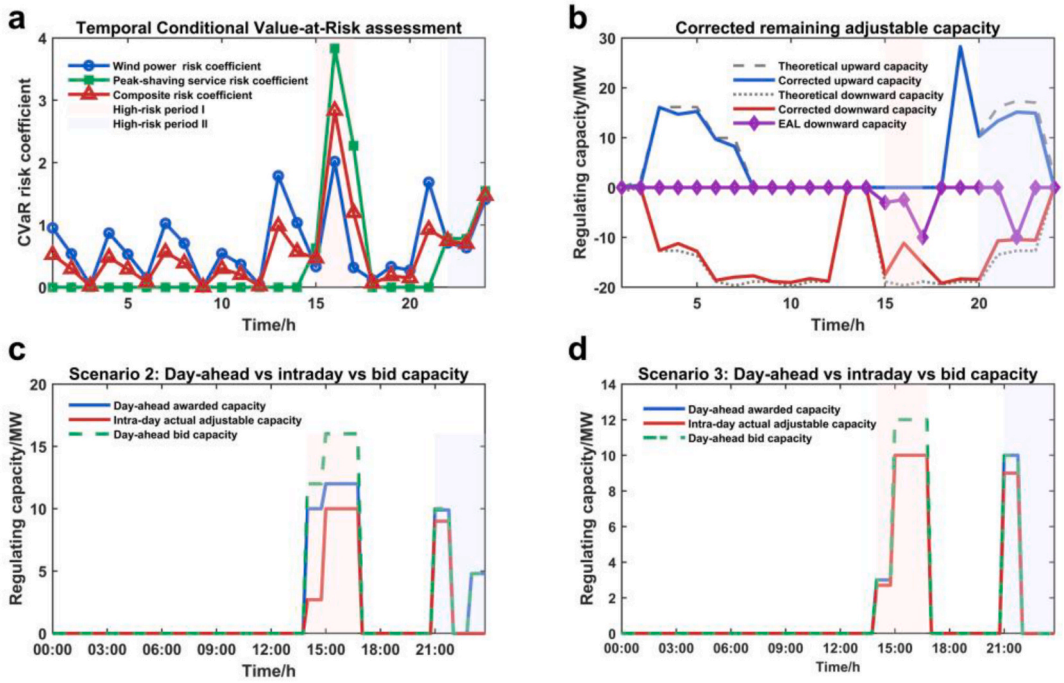


Fig. 14. Temporal CVaR risk assessment analysis.

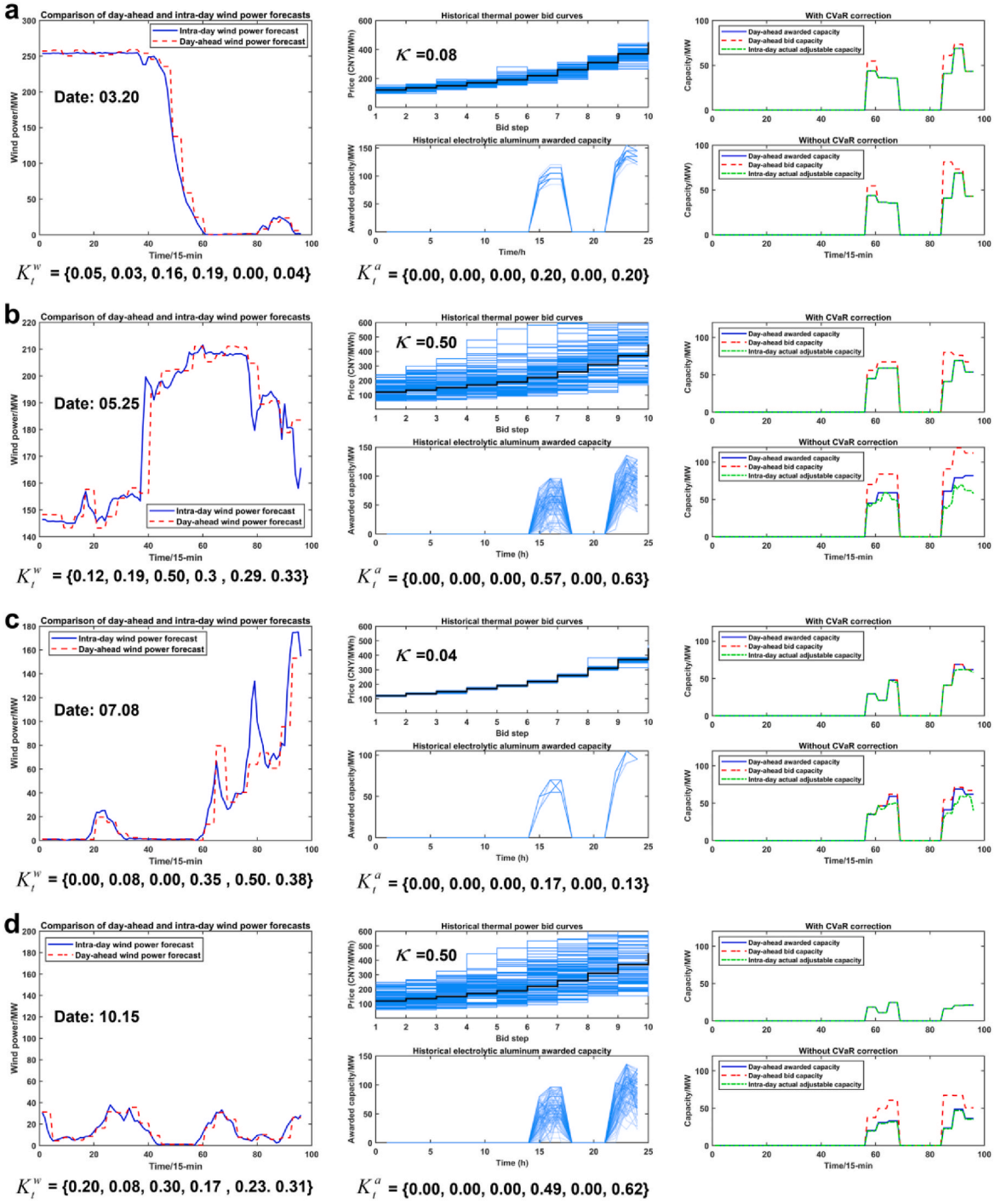


Fig. 15. Capacity correction effectiveness under multiple scenarios.

These results demonstrate that the proposed temporal CVaR-based peak-shaving capacity correction model effectively reduces capacity default risk under different uncertainty scenarios, with more prominent effects under severe conditions, thereby validating its adaptability and effectiveness.

3.4.2. Correlation analysis

Based on the data of scenarios a–d in Fig. 15, the correlations of K_t^w and K_t^a with the two types of regulation capacity gaps are analyzed, as shown in Fig. 16. K_t^w is positively correlated with the gap between P_t^{awd} and P_t^{act} ($r = 0.545$). The main reason is that the multi-process coordinated regulation of the EAL smooths out intra-day wind power fluctuations, thereby partially mitigating the propagation of deviations into the capacity gap. In contrast, K_t^a is strongly positively correlated with the gap between P_t^{bid} and P_t^{awd} ($r = 0.805$).

Despite the limited scenario set and discrete data points, the underlying correlations remain evident; a broader range of scenarios would yield a more intuitive representation.

Furthermore, while the parametric modeling of rivals' bids in the peak-shaving market cannot precisely replicate real-world clearing outcomes, such simplification is justifiable within the scope of this research. Future efforts to simulate authentic market dynamics should incorporate advanced methodologies—such as evolutionary game theory [44] or deep reinforcement learning [45]—to further enhance model fidelity.

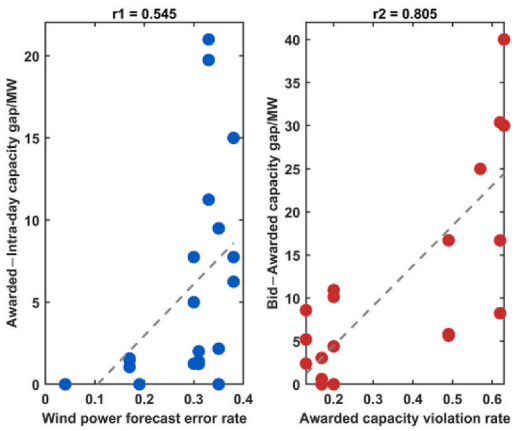


Fig. 16. Correlation scatter plots.

Table 6
Scenario day configurations for three long-term distributions.

Distribution type	Scenario a (days)	Scenario b (days)	Scenario c (days)	Scenario d (days)
Stable	16	3	7	4
Normal	13	4	8	5
Severe	10	5	9	6

3.4.3. *Long-term economic analysis*

Based on the four scenarios established in Fig. 15, three long-term distributions (see Table 6) reflecting different operating environments are defined to evaluate the long-term benefits of the proposed temporal CVaR correction model. The corresponding clearing prices for each scenario are given in Appendix B. Table B4.

The penalty for a capacity deficit is calculated as the penalty multiplier M multiplied by the clearing price and the energy deficit, with M set to 2, 3, and 5 for sensitivity analysis.

As shown in Fig. 17, under the stable distribution, the difference in monthly net profit before and after correction is small, indicating that the temporal CVaR correction does not impair economic performance under normal operating conditions. Under the normal and severe distributions, the monthly net profit of the corrected strategy exceeds that of the uncorrected expected-value optimization, and the advantage becomes more pronounced as M increases. Therefore, for an electrolytic aluminum industrial park pursuing long-term robust operation, the temporal CVaR correction offers better comprehensive economic benefits than expected-value optimization.

4. Conclusion

We propose a multi-timescale optimal scheduling method for electrolytic aluminum industrial parks, accounting for the heterogeneity of multi-process systems and the dual risks inherent in transmission and distribution networks. To address the dual internal (captive wind power) and external (peak-shaving market price) uncertainties, the method leverages the heterogeneous regulation capabilities of the three core production processes—electrolysis, feeding, and aluminum tapping. This enables a high utilization of local wind power while providing reliable peak-shaving services to the grid. The case study leads to the following conclusions:

Compared to the traditional approach that regulates only the electrolysis process, the proposed refined multi-process regulation model achieves functional complementarity by utilizing the distinct characteristics of each process. This approach overcomes the regulation limitations of the sole electrolysis process and suppresses short-term wind power fluctuations that are untrackable by the electrolysis process due to its thermal inertia. Consequently, it increases the wind power consumption rate of the system by approximately 9.83%, reduces the coal consumption cost of thermal power units by 65.1×10^3 CNY, and raises aluminum product revenue by 168.9×10^3 CNY.

The proposed temporal CVaR peak-shaving capacity correction model achieves a coordinated balance between economy and reliability. In intra-day scheduling, the peak-shaving revenue drops by only 200 CNY from the day-ahead value—much smaller than the drop in the uncorrected scenario — effectively avoiding ancillary service default risk induced by internal and external dual uncertainties. Multi-scenario analysis results show that under different wind power output fluctuations and market price uncertainties, the model significantly reduces capacity default risk, particularly in highly uncertain scenarios.

Building on the aforementioned models, the multi-timescale optimal scheduling strategy coordinates wind power consumption with ancillary peak-shaving by allocating the heterogeneous production processes to the day-ahead, intra-day, and real-time scheduling stages. This leads to a coordinated operational mode: "the electrolysis process handles bulk power shifting, the aluminum tapping process provides discrete supplementation, and the feeding process performs rapid fine-tuning." The results demonstrate that this strategy not only keeps key production parameters (e.g., electrolyte temperature and alumina concentration) within safe operating ranges but also maintains a wind power consumption rate of approximately 93% and a peak-shaving revenue of 11.8×10^3 CNY across multiple timescales.

CRediT authorship contribution statement

Yulong Yang: Writing – original draft, Visualization, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

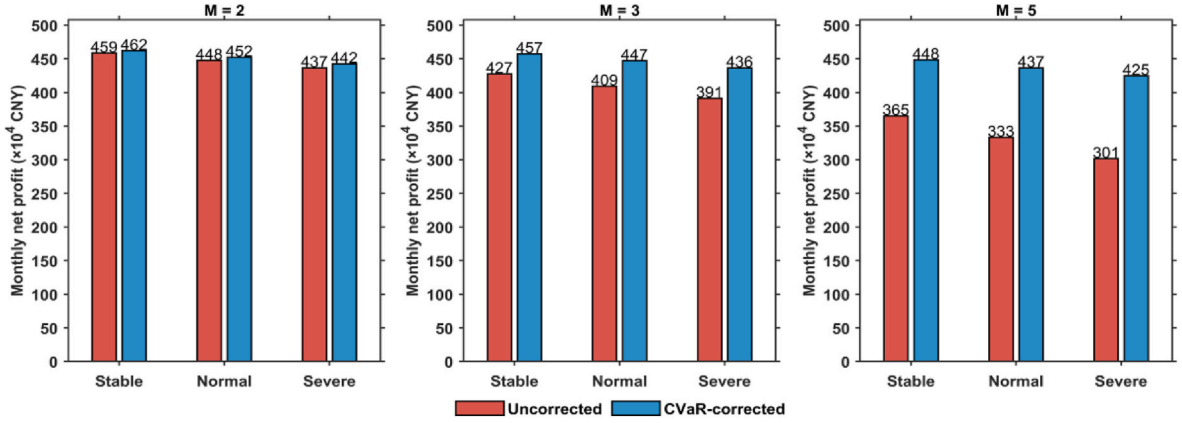


Fig. 17. Comparison of monthly net profit under different distributions and penalty multipliers.

Songnan Wang: Visualization, Resources, Formal analysis, Data curation. **Gangui Yan:** Writing – review & editing, Supervision, Project administration. **Gang Mu:** Supervision. **Zhe Chen:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the Key Research and Development Program Project of Xinjiang Uyghur Autonomous Region (2022B01020-1)

Appendix A. Scheduling plans for different scenarios

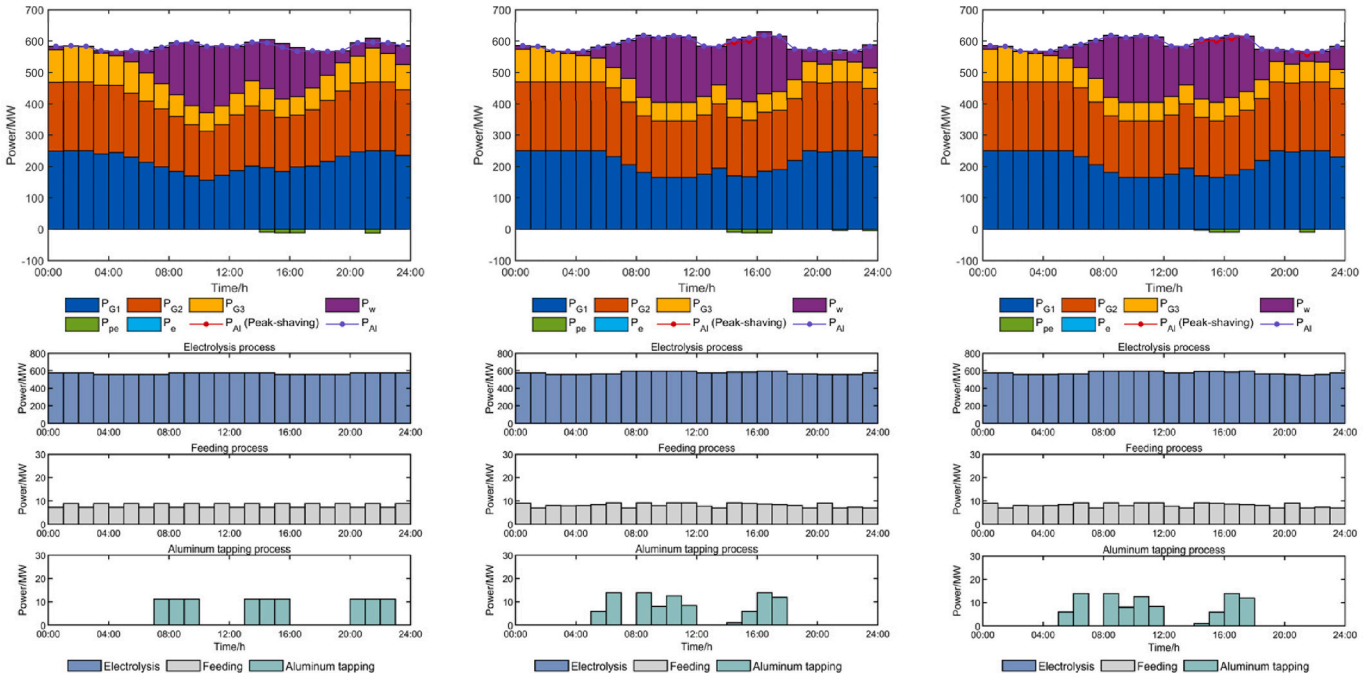


Fig. A1. Day-ahead scheduling plans for different scenarios

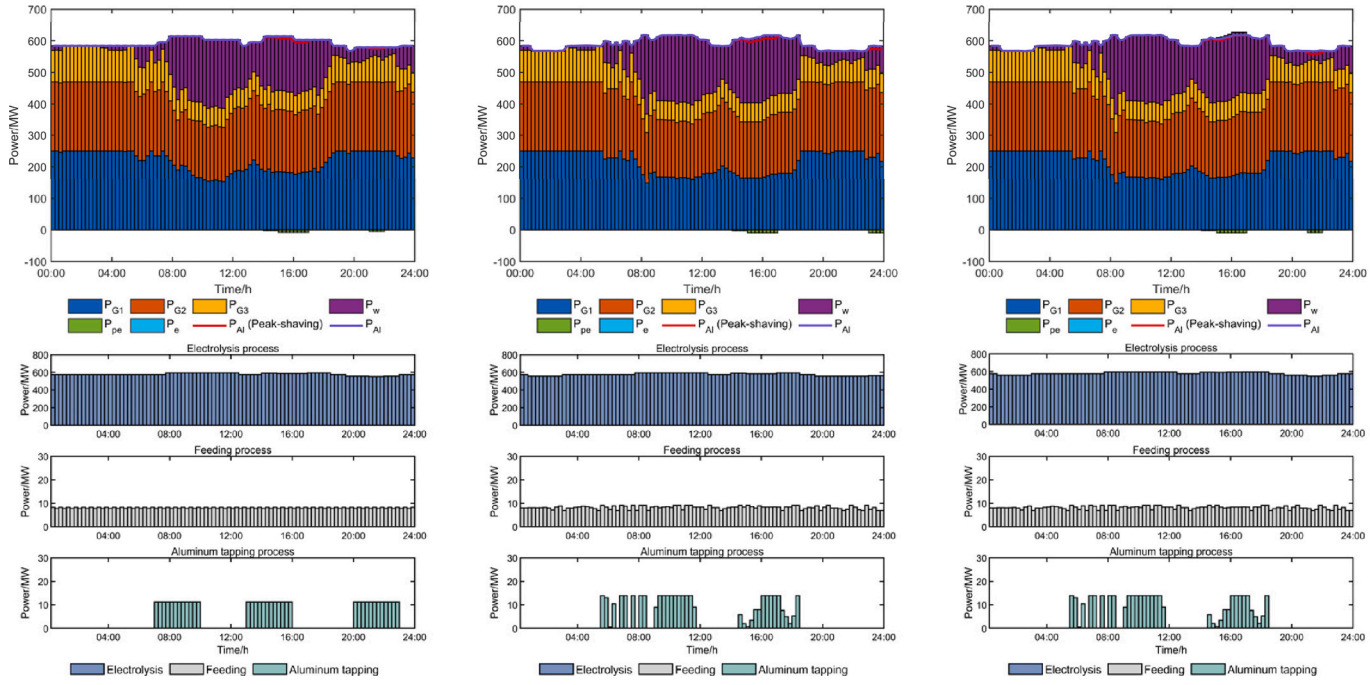


Fig. A2. Intra-day scheduling plans for different scenarios

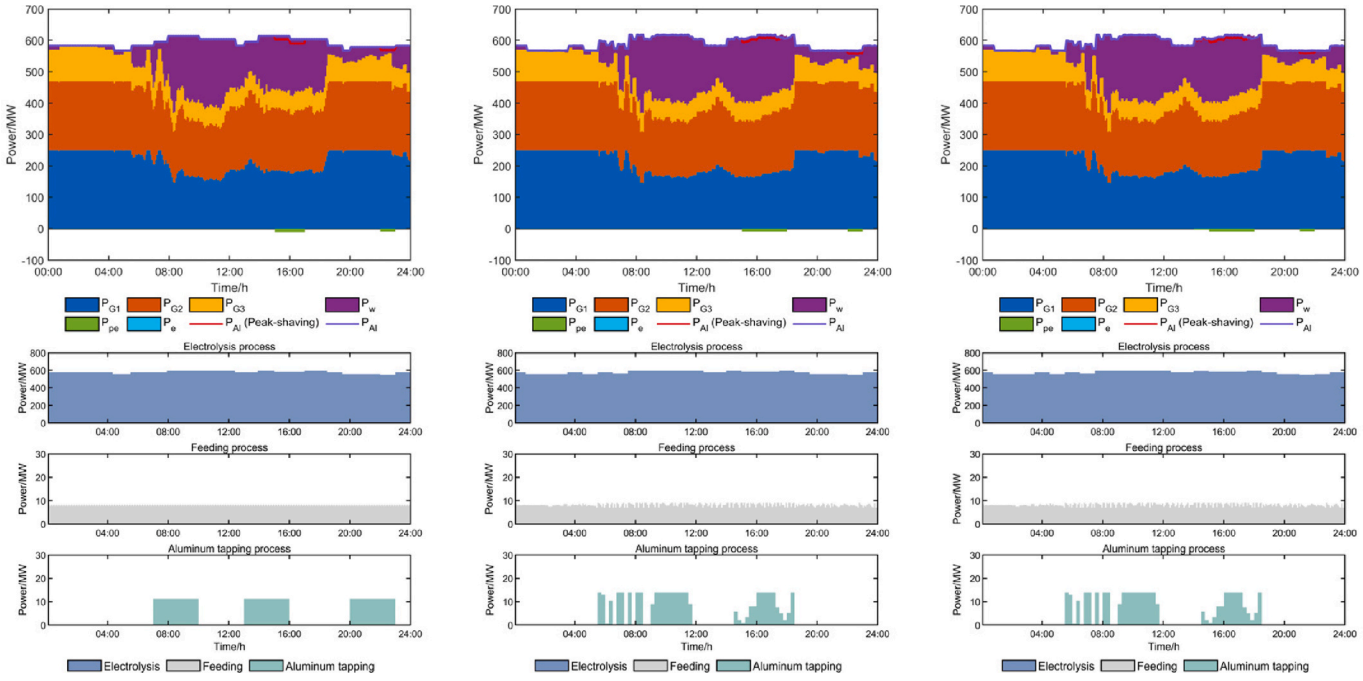


Fig. A3. Real-time scheduling plans for different scenarios

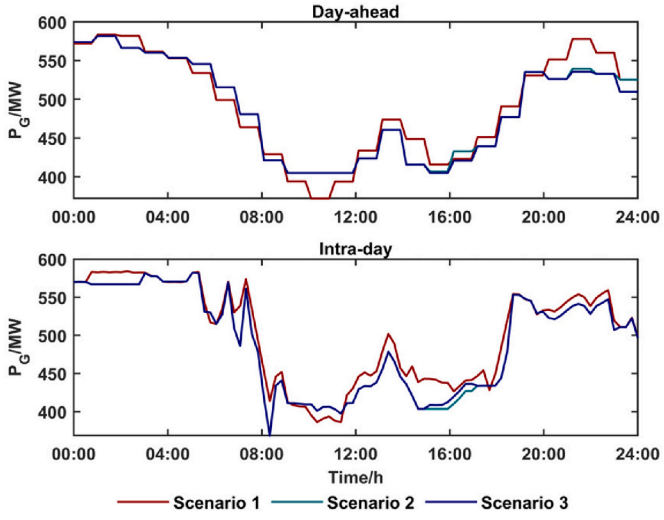


Fig. A4. Comparison of output performance of captive thermal power units in different scenarios

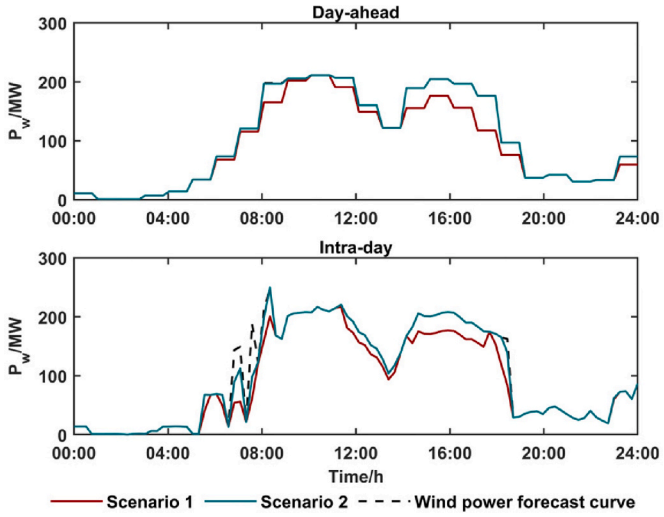


Fig. A5. Comparison of captive wind power consumption in different scenarios

Appendix B. Detailed parameters

Table B1
Parameters of thermal power units

thermal power units	P^G_{max}/ MW	P^G_{min}/ MW	C_{st}/ CNY	$a/b/c$/ (CNY /MW^2)/(CNY /MW)/(CNY)	P^G_{up}, P^G_{down}/ (MW /min)
G1	250	125	3850	0.00048/16.2/1000	30/25
G2	220	115	3920	0.00031/17.3/970	25/20
G3	180	60	3600	0.0002/16.6/700	20/15

Table B2
Peak-shaving market demand and benchmark compensation price [46].

Scenarios	15:00	16:00	17:00	22:00	23:00	24:00
Market benchmark offer (CNY /MWh)	350	360	360	370	410	400
Peak-shaving capacity (MW)	10	12	12	13	17	16

Table B3
Stepwise bid prices and upward capacities of the thermal power unit

Step	1	2	3	4	5	6	7	8	9	10
Compensation price (CNY/MWh)	120	135	150	170	190	220	260	310	370	450
Upward capacity (MW)	5	10	10	10	15	15	15	20	22	25

Table B4
Clearing prices for Scenarios a–d during the six periods (CNY/MWh)

Scenario	15h	16h	17h	22h	23h	24h
a	172	220	220	218	220	310
b	166	166	166	210	210	220
c	190	220	190	210	202	210
d	220	260	220	260	310	310

Appendix C. Supplementary Mathematical Formulations and Derivation

C.1. Electro-thermal coupling calculation formulas for the electrolytic cell shell

$$Q_{sur1} = \left(Q_t^{conv1} + Q_t^{rad1}\right) dt \tag{C.1}$$

$$Q_t^{conv1} = A_1 h_{conv1} \left(T_t^s - T_t^0\right) \tag{C.2}$$

$$Q_t^{rad1} = A_1 q_t^{rad1} = A_1 \varepsilon C_0 \varphi \left[\left(\frac{T_t^s + 273}{100} \right)^4 - \left(\frac{T_t^0 + 273}{100} \right)^4 \right] \tag{C.3}$$

where Q_{sur1} represent the heat dissipation from the shell to the ambient air, Q_t^{conv1} and Q_t^{conv1} represent the convective and radiative heat losses between the shell and the ambient air respectively.

C.2. Standard mathematical model of CVaR

The conditional value-at-risk V^{CVaR} at the confidence level ξ is given by:

$$V^{CVaR} = \frac{1}{1-\xi} \int_{f(x,y) \geq V^{VaR}} f(x,y) \rho(y) dy \tag{C.4}$$

$$\begin{cases} V^{VaR} = \min\{a \in \mathbf{R} : F(x,a) \geq \xi\} \\ F(x,a) = \int_{f(x,y) \leq a} \rho(y) dy \end{cases} \tag{C.5}$$

where x represents the decision variable; v represents the random variable; V^{VaR} represents the value-at-risk; $F(x,a)$ represents the distribution function for which the risk function $f(x,y)$ does not exceed the threshold a; $\rho(y)$ represents the probability density function of the random variable y; q represents the number of samples.

C.3. Thermal power unit/wind power output constraints

1) Thermal power unit output constraints

Output upper and lower limit constraints:

$$s_{i,t} P_{\min}^G \leq P_{i,t}^G \leq s_{i,t} P_{\max}^G \tag{C.6}$$

Power ramp constraints:

$$s_{i,t+1} P_{i,t+1}^G - s_{i,t} P_{i,t}^G \leq P_{up}^G \tag{C.7}$$

$$s_{i,t} P_{i,t}^G - s_{i,t+1} P_{i,t+1}^G \leq P_{down}^G \tag{C.8}$$

Positive and negative spinning reserve constraints:

$$P_{\max}^G - P_t^G \geq a_{Al} P_t^{Al} + a_w P_t^w \tag{C.9}$$

$$P_t^G - P_{\min}^G \geq a_{Al} P_t^{Al} + a_w P_t^w \tag{C.10}$$

where, a_{Al} and a_w represent the spinning reserve coefficients for the EAL and wind power forecast, taken as 0.05 and 0.15 respectively [36].

2) Wind power output constraints

$$0 \leq P_t^w \leq P_t^{wf} \tag{C.11}$$

Data availability

Data will be made available on request.

References

[1] National Energy Administration. Implementation plan for optimizing power System regulation capacity enhancement action (2025–2027) [EB/OL]. Beijing: National Energy Administration; 2025.

[2] National New Energy Consumption Monitoring and Early Warning Center. National new energy grid integration and consumption data. June 2025 [EB/OL]. (2025-08-11).

[3] Lund PD, Lindgren J, Mikkola J, et al. Review of energy system flexibility measures to enable high levels of variable renewable electricity. Renew Sustain Energy Rev 2015;45:785–807.

[4] Fu Y, Yang H, Yang X, et al. A holistic picture of the carbon emission responsibility in China's aluminium supply chain: Production-side flow analyses, consumption-side responsibility allocation, and driving factor analysis. Energy 2025:136372.

[5] Zhu X, Jin Q. Exploring the carbon neutrality pathway for China's aluminium industry: an analysis from 1950 to 2060. Appl Energy 2025;393:126090.

[6] Han Z, Li Z, Wang W, et al. Multi-time optimization scheduling strategy for integrated energy systems considering multiple controllable loads and carbon capture plants. Energies 2024;17(23):5995.
[7] Yang Y, Li S, Wang S, et al. Low-Carbon economic dispatch method for integrated energy in aluminum electrolysis considering production safety constraints. Processes 2025;13(11):3442.
[8] Yang Y, Yan H, Wang J, et al. System optimization scheduling considering the full process of electrolytic aluminum production and the integration of thermal power and energy storage. Energies 2025;18(3):598.
[9] Yue X, Liao S, Xu J, et al. Collaborative optimization of renewable energy power systems integrating electrolytic aluminum load regulation and thermal power deep peak shaving. Appl Energy 2024;373:123869.
[10] Bao Y. Study on the strategy of high energy-consuming electrolytic aluminum load participating in power system frequency modulation and auxiliary service [D]. Wuhan University; 2019.
[11] Lin S, He T, Shen Y, et al. Bilevel optimal dispatch model for a peak regulation ancillary service in an industrial park of energy-intensive loads. Elec Power Syst Res 2024;230:110272.
[12] Yu Q, Xu J, Liao S, et al. Adaptive load control of electrolytic aluminum for power system frequency regulation based on the aluminum production operation state. Energy Rep 2022;8:1259–69.
[13] Liao S, Pi S, Xu J, et al. Emergency control strategy for electrolytic aluminum load considering the load's transient overcurrent characteristics and transient voltage stability constraints. IEEE Trans Power Syst 2025;40(2):1699–711.
[14] Tabereaux AT, Peterson RD. Aluminum production[M]//Treatise on process metallurgy. Elsevier; 2024. p. 625–76.
[15] Li J, Feng D. Intelligent feeding control strategy based on aluminum concentration identification in aluminum electrolysis. Light Met 2019;2:31–6.
[16] Li J, Hi Y, Nan D. Scheduling optimization of electrolytic aluminum collaborative aluminum allocation under regional agglomeration production mode. Nonferrous Metals Engineering 2022:63–9.
[17] Wu Z, Kang J, Mosteiro-Romero M, et al. A distributionally robust optimization model for building-integrated photovoltaic system expansion planning under demand and irradiance uncertainties. Appl Energy 2024;372:123740.
[18] Yang M, Zhu Y, Yu X, et al. Distributionally robust low-carbon scheduling of integrated energy system considering source—load collaborative carbon reduction under multiple time scales. Electric Power Automation Equipment 2025;45(2):34–42.
[19] Ma M, Long Z, Liu X, et al. Distributionally robust optimization of electric–thermal–hydrogen integrated energy system considering source–load uncertainty. Energy 2025;316:134568.
[20] Mu Y, Wang C, Cao Y, et al. A CVaR-based risk assessment method for park-level integrated energy system considering the uncertainties and correlation of energy prices. Energy 2022;247:123549.
[21] Zhang Q, Jin H, Wu X, et al. Two-stage optimal scheduling strategy for electric-hydrogen integrated energy stations considering demand response in energy and reserve services markets. Energy 2025:137697.
[22] Zhu F, Zhao L, Liu W, et al. A stochastic optimization framework for short-term peak shaving in hydro-wind-solar hybrid renewable energy systems under source-load dual uncertainties. Appl Energy 2025;400:126597.
[23] Zang Y, Cui J, Wang S, et al. Low-carbon economic optimization strategy for industrial loads in parks considering source-load-price multivariate uncertainty. Elec Power Syst Res 2025;239:111200.
[24] Wan J, Yao K, Ren G, et al. Spatiotemporal diurnal modulation characteristic of wind speed and power generation revealed by its measured data processing. Comput Intell Neurosci 2022;2022(1):5722770.
[25] Gong T, Che J, Wang B, et al. Short-term wind power prediction correction method considering error probability distribution and fluctuation characteristics. High Volt Eng 2025;51(1):379–89.
[26] Wang P, Wang Y, Lin Z, et al. Optimal decision in spot-deep peak regulation market for thermal power suppliers considering uncertainty. Electr Power Autom Equip 2024;44:208–16.
[27] Tang D, Ge P, Yuan C, et al. Optimal management of coupled hydrogen-electricity energy systems at ports by multi-time scale scheduling. Appl Energy 2025;391:125885.
[28] Jiang H, Liu X, Zhou H, et al. Multi-time-scale optimal scheduling strategy of electricity-heat-cold-gas integrated energy system considering ladder carbon trading. Energy Rep 2025;13:4000–14.
[29] Chu S, Ge WS, Teng Y, et al. Multi-time-scale scheduling strategy for combined accommodation of wind power by seawater desalination and electric heating loads. Autom Electr Power Syst 2023;47:1–16.
[30] Wang X, Wang H, Fan T, et al. Optimal scheduling of source-load-storage at multiple time scales considering the participation of multiple industrial loads. Sci Technol Eng 2024;24(15):6290–9.
[31] Yang Y, Yan G, Mu G, et al. Hierarchical bidding strategy for heterogeneous P2H loads and wind power: state-driven aggregation and switching time scheduling. Energy 2025:137766.
[32] Li L, Zhang J, Chen Y, et al. Multi-Process, energy-intensive industrial loads participation in demand side response power control strategies. 2024 6th Asia energy and electrical engineering symposium (AEEES). 2024. p. 259–64.
[33] Liu R, Liu X, Tang J, et al. Multi-timescale dispatching method for industrial microgrid considering electrolytic aluminum load characteristics. Processes 2025;13(5):1411.
[34] Yang S, Zhang HL, Zou Z, et al. Calculation of heat transfer coefficient between aluminum reduction cell surface and surroundings. Chi J Nonferr Metals 2015;25:515–22.
[35] Yang S. An interfacial heat transfer mechanism based integrated model and its application for thermal field analysis in aluminium Reduction Cells[D]. Changsha: Central South University; 2013. p. 58–61.
[36] Yue X, Liao S, Fu L, et al. Hierarchical optimization strategy of high-penetration wind power system considering electrolytic aluminum energy-intensive load regulation and deep peak shaving of thermal power. Power Syst Technol 2024;48:3186–96.
[37] Hestetun K, Hovd M. Detection of abnormal alumina feed rate in aluminium electrolysis cells using state and parameter estimation. In: Computer aided chemical engineering, vol. 21. Elsevier; 2006. p. 1557–62.
[38] Li S. Research on key technologies of intelligent control of the whole process of electrolytic aluminum production. Lanzhou University of Technology; 2020 [D].
[39] National Energy Administration Xinjiang Office. Xinjiang electricity ancillary service market implementation rules [EB/OL]. 2025-07-09 [2026-04-19].
[40] Savelli I, Giannitrapani A, Paoletti S, et al. An optimization model for the electricity market clearing problem with uniform purchase price and zonal selling prices. IEEE Trans Power Syst 2017;33(3):2864–73.
[41] National Development and Reform Commission, National Energy Administration. Electricity load management approach. 2023 Edition 2023-09-27.
[42] Yang Z, Chen L, Hao L, et al. Industrial load peak regulation and its secondary frequency regulation capacity assessment considering intra-production segment characteristics. Power Syst Technol 2025;49:3613–22.
[43] Ren J, Xue C, Ma X, et al. Two-stage model of peak regulation ancillary service market with source-load interaction. Autom Electr Power Syst 2021;45(18):94–102.
[44] Cheng L, Huang P, Zou T, et al. Evolutionary game-theoretical approaches for long-term strategic bidding among diverse stakeholders in large-scale and local power markets: basic concept, modelling review, and future vision. Int J Electr Power Energy Syst 2025;166:110589.
[45] Hu F, Zhao Y, Yu Y, et al. Strategic bidding with price-quantity pairs based on deep reinforcement learning considering competitors' behaviors. Appl Energy 2025;391:125874.
[46] Wang H, Zhou Z, Li X. Multi-time scale optimal scheduling of virtual power plant considering electricity energy-peaking markets. In: Proceedings of the CSU-EPSA; 2025. p. 1–10.